TAMIL NADU AGRICULTURAL UNIVERSITY

PROCEEDINGS

43rd Rice Scientists Meet 2024

(15th July, 2024)

Lead Centre & Venue

Tamil Nadu Rice Research Institute
Aduthurai - 612 101

Directorate of Research

Tamil Nadu Agricultural University
Coimbatore - 641 003

2024

PROCEEDINGS
43rd Rice Scientists Meet
(15th July, 2024)

The 43rd Rice Scientists Meet was held on 15th July, 2024, at Tamil Nadu Rice Research Institute, Aduthurai. Pre-review of individuals research projects, action taken on OFT, action plan projects were made by concerned Technical Directors in the concurrent session at TNAU, Coimbatore. **Dr. V. Geethalakshmi,** Vice-Chancellor chaired the meeting and emphasized on the popularization of newly released TNAU rice varieties for more horizontal spread. Madam insisted mainly on widening the genetic base, development of improved landraces, DSR management practices, sustainable soil fertility enhancement, IPM for pests and diseases, varietal resistance to the changing pest scenario. It was suggested to identify package of practices for direct seeded rice, mechanization in rice cultivation, bio-intensive pest and diseases management. **Dr. N. Senthil**, Director (CPMB&B), TNAU welcomed the gathering.

The Action taken reports of 42nd Rice Scientists Meet 2023, Research Highlights and Action plan for 2023-24 was presented by **Dr. R. Ravikesavan**, Director (CPBG), **Dr. M.K. Kalarani**, Director (Crop Management), **Dr. P. Balasubramaniam**, Director (NRM) and **Dr. M. Shanthi**, Director (CPPS) for Crop Improvement, Crop Management, Natural Resource Management and Crop Protection respectively. **Dr. K. Subrahmaniyan,** Director, TRRI, Aduthurai proposed a formal vote of thanks.

The Proceedings of the 43rd Rice Scientist Meet are furnished below in the following headings:

I. CROP IMPROVEMENT
- A. Cultures recommended for variety release/OFT/ART/MLT
- B. Action plan
- C. Project wise remarks

II. CROP MANAGEMENT
- A. For Adoption/OFT/Information
- B. Action plan
- C. Project-wise Remarks

III. NATURAL RESOURCE MANAGEMENT
- A. Technologies for adoption/OFT/Information
- B. Action plan
- C. Project wise remarks

IV. CROP PROTECTION
- A. Action plan
- B. Project wise remark

V. REMARKS

VI. LIST OF PARTICIPANTS

I. CROP IMPROVEMENT

A. CULTURES RECOMMENDED FOR VARIETY RELEASE/ART/OFT/MLT

I. Cultures recommended for submission of variety release proposal to UVTRSC (After finalization of ART results only)

1. AD 13253–Medium- Irrigated

Parentage	:	**AD 13253** (AD 01246 / CO(R) 49)
Duration	:	132 days
Average yield	:	MLT (15): 5414 kg/ha which is 15.3% increase over ADT 39 ART (90): 5815 kg/ha which is on par with ADT 39.
Salient Features	:	Moderate Resistance to BLB under Artificial condition, Medium Slender grain with good cooking properties

II. Cultures identified for On Farm Trials (2024-25)

1. TR 15057

Parentage	:	**TR 15057** (IR 20 / CSR 23 // CSR 23)
Duration	:	115 days
Average yield	:	4659 kg/ha in 115 days (13.4 % over ADT 53, 13.9% over TRY 2)
Salient Features	:	Salinity tolerant, moderately resistant to BLB, sheath rot, Brown spot

III. Cultures identified for Second Year ART & OFT (2024-25)

Rice 10/2024 -25: Rainfed- Early (Sept.- Oct.)			
Cultures with Parentage and checks	**Yield and duration**	**Special attributes**	**Locations**
PM 17011 (ASD 16 / Chandaikar) Check: TKM 15 & CO 53	4400 kg / ha & 105-110 days	Short duration; Drought Tolerant; Suitable for direct seeding under rainfed and semi dry condition during NEM period (Sep – Oct)	Ramanathapuram, Sivagangai and Virudhunagar Districts

Cultures with Parentage and checks	**Yield and duration**	**Special attributes**	**Locations**
Rice 15/2024-25 Special transplanted (Sep - Oct) sowing: 125-140 (days) Medium Duration (2024-25)			
CBMAS14065 (I. White ponni/ Apo) Check: BPT 5204, TKM 13	6186 kg / ha & 130-135 days	14.5% over BPT 5204 Super fine rice, Good cooking quality HRR (62.1%), High LER MR - SB & RTD	All Districts except Virudhunagar, Ramnad, Sivagangai and The Nilgiris

IV. Cultures identified for ART & OFT under fast-track mode (2024-25)

Cultures with Parentage and checks	Yield and duration	Special attributes	Locations
Rice 15/2018-19 Special transplanted (Sep - Oct) sowing: 125-140 (days) Medium Duration – Super Fine (2024-25)			
AD 18559 (Kalajoha / IW.Ponni) Checks: RNR 15048	5619 kg/ha in 135 days (8.0 % over BPT 5204)	Excellent cooking quality similar to RNR 15048	All Districts except Virudhunagar, Ramnad, Sivagangai and The Nilgiris

V. Cultures identified for first year of ART & OFT (2024-25)

ART Rice 3- Transplanted (*Sornavari/Kar/Kuruvai*, April-July) 105-115 days			
Cultures with Parentage and checks	**Yield and duration**	**Special attributes**	**Locations**
AD 18028 (AD 08010/AD 07073) Check: ADT 53	6420 kg/ha in 115 days 9.75, 14.5 and 8.18 % yield superiority over the check entries ADT 53, ADT 57 and CO 54	Recorded a high LER of 1.50 and lowest BER of 1.37 with good volume expansion (VER: 4.0) soft gel (GC: 105) and intermediate amylose content (22.1%) recorded 65 percent HRR.	All Districts except Virudhunagar, and The Nilgiris
AD16028 (ADT 56) Check: **CO 55**	6086 kg/ha in 115 days	Fine grains type registered high milling (69.4%) and head rice yield (58.8%). It possesses intermediate amylose (23.8%) and good cooking and eating qualities.	

Rice 4/2024-25: Transplanted Mid Early (Oct.25th –Nov 10th, Sowing) - Fine grain			
Cultures with Parentage and checks	**Yield and duration**	**Special attributes**	**Locations**
CB 19127* (CR 1009/BG 358) Check- RNR 15048	6106 kg/ha kg/ha in 125 days which is 15.03 % higher than the RNR 15048.	The HRR recorded is 60% for this culture. It possesses fine slender grain with LER 1·60, VER (4·4) and with Intermediate Cooking temperature (ASV 3) and produces soft gel consistency GC 105	All Districts except Virudhunagar, Ramnad, Sivagangai and The Nilgiris

ART Rice 6/2024-25-Transplanted (August 15 – September 10 sowing) 140 days and above			
Cultures with Parentage and checks	**Yield and duration**	**Special attributes**	**Locations**
AD 18145 (CR 1009/MTU 1075) Check: CR 1009	6350 kg/ha in 145 days	Bold grains culture AD 18145 possessed preferable quality traits such as HRR of 65 percent with LER 1.96, VER 4.9, soft gel consistency, and intermediate amylose content (23.4%).	All Districts except Virudhunagar, Ramnad, Sivagangai and The Nilgiris
ART - (Direct sowing Herbicide trial)- 2024-25			
Cultures with Parentage and checks	**Yield and Duration**	**Special Attributes**	**Locations**
CBHTR 22001 CBHTR 22002 Check: CO 51	5835 kg/ha 110 days 5456 kg/ha 110 days	Herbicide Tolerant Trait Medium Slender grain type Suitable for Direct sowing	Wet DSR: Thanjavur, Thiruvannamalai, Thiruvallaur, Erode, Tiruppur Dry DSR: Thiruvarur, Nagapattinam, Ramnad, Sivagangai, Mayiladuthurai

VI. Multilocation Trials (2024-25)

Table 1. MLT- I (100 – 120 days maturity; May – June Sowing)

S. No.	Designation	Cross Combination	Maturity	Rice Grade	Yield (Kg/ha)	Nominating centres
New						
1.	AD 21061*	ADT 43/WGL 32100	115	SF	6750	ADT
2.	AD 22186*	IET 22075/RNR 15048	115	SF	6600	ADT
3.	AD 22143*	AD 14098/AD 11011	115	SF	6733	ADT
4.	TM 15134	ADT 43/ NLR 34449	115	MS	6635	TKM
5.	CB 21515	CO 51 / CO 54	116	MS	6882	CBE
6.	CB 21533	CO 51/PRNP 9108	117	MS	6888	CBE
7.	TNTRH 61	TNAU 45S / CBSN 405	115	MB	8725	CBE
8.	ACM 20016	I.W.Ponni/ CO 43	117	SB	6720	MDU
9.	AS 20026	ASD 16 / Manjal saradai	118	SB	6825	ASD
Checks	:	ADT 57, CO 55*, CORH 3 and TPS 5				
Replications	:	Three				
Plot size	:	9 m^2				
Spacing	:	15 x 10 cm				

Locations (11)	:	Aduthurai, Coimbatore, Ambasamudram, Tirur, Thirupathisaram, Madurai, Killikulam, Thanjavur, Paiyur, Cuddalore and Vaigai Dam
Seed despatch	:	5.0 kg to be sent before June 20th

***Will be compared with superfine check variety CO 55**

Table 2. MLT- II (121 – 130 days maturity; September-October Sowing)

S. No.	Designation	Cross combination	Maturity	Rice grade	Yield (Kg/ha)	Nominating centres
Repeat						
1.	CB 17537	CO 51/ WGL 536	124	MS	6532	CBE
2.	ACK 20026	White Ponni / ACK 12024	128	MS	6707	KKM
3.	TM 20125	ADT 37 / FL 478	125	MS	6493	TKM
New						
4.	AD 21198*	Turantdhan / IET 22075	123	SF	6309	ADT
5.	AD20516*	TKM13/AD15115	125	SF	6145	ADT
6.	AD20619*	Jaishriram/Jeeraga samba	125	SF	5916	ADT
7.	AD 21188*	Turant dhan /IET 22075	125	SF	6350	ADT
8.	TM 15118	BPT 5204/ CB 01001	125	MS	6568	TKM
9.	CB 22915*	ADT 43*4 /CT 9993	125	SF	6416	CBE
10.	CB 21 520	CO 51 / RNR 17494	126	MS	6632	CBE
11.	CB 21 531	CO 51 / PRNP 9108	126	MS	6592	CBE
12.	CB 22166*	CO 56 / Kanagavalli Sona	126	SF	6118	CBE
13.	TNTRH 107	TNAU 147S / CBSN 403	128	MB	8131	CBE

Checks	:	TKM 13, *RNR 15048 (for SS cultures), CORH 5
Replications	:	Three
Plot size	:	9 m^2
Spacing	:	15 x 10 cm
Locations (11)	:	Aduthurai, Coimbatore, Madurai, Ambasamudram, Tirur, Thirupathisaram, Killikulam, Thanjavur, Paiyur, Cuddalore and Vaigai Dam
Seed despatch	:	5.0 kg to be sent before June 20th

***Will be compared with superfine check variety RNR 15048**

Table 3. MLT- III (131 – 140 days maturity; September-October Sowing)

S. No.	Designation	Cross combination	Maturity	Rice Grade	Yield (Kg/ha)	Nominating centres
1.	AD 19513*	AD07302/AD08142	135	SF	6420	ADT
2.	AD 20749*	Jaishriram/Jeeraga Samba	135	SF	6403	ADT
3.	AD19661	CO50 /AD07302	135	MS	6720	ADT
4.	TM 15683	ADT 46/ WGL 14	134	MS	6630	TKM
5.	CB 21112*	CO 56 / Sriramsena	130	SF	6485	CBE
6.	CB 22158*	CO 56 / BPT 5204	131	SF	6225	CBE

7.	CB 22138	IW Ponni / CO 52	132	MS	6435	CBE
8.	CB 22121	CO 52 / IET 23750	138	MS	6437	CBE
9.	TNTRH 81	TNAU 60S / CBSN 177	132	MB	10180	CBE
10.	TNTRH 151	TNAU 147S / CBSN 386	135	MS	8390	CBE
11.	ACM 20045	RNR 15048/ CB13543	130	MS	6393	MDU

Checks	:	ADT 54, CO 56, TPS 3, CORH 4, BPT 2782*
Replications	:	Three
Plot size	:	9 m^2
Spacing	:	20 x 10 cm
Locations (12)	:	Aduthurai, Coimbatore, Ambasamudram, Tirur, Thirupathisaram, Sirugamani, Madurai, Killikulam, Thanjavur, Paiyur, Cuddalore & Vaigai Dam
Seed despatch	:	5.0 kg to be sent before

***Will be compared with superfine check variety BPT 2782**

Table 4. MLT- IV (> 140 days maturity; August sowing)

S. No.	Designation	Cross combination	Maturity	Rice grade	Yield (Kg/ha)	Nominating centres
Repeat						
1.	AD 20444	CR 1009/OR 23780-1	148	SB	7097	ADT
2.	AD 20389	CR 1009/MTU1075	152	SB	6499	ADT
3.	AD 20420	CR 1009/MTU1075	150	MS	6331	ADT
4.	ACK 20011	Mutant of CR 1009	145	MS	6852	KKM
New						
5.	AD 22411*	AD13116 / BPT 2270	145	SS	6449	ADT
6.	AD 22455	AD13116 / BPT 2270	148	MS	6505	ADT
7.	AD 21307	CO50/WGL32100//CR1009/BPT2270	150	MS	6605	ADT
8.	AD 18143	CR 1009/ MTU1075	155	SB	7053	ADT
9.	ACK 20024	Gamma ray mutant of CR 1009	145	MS	6707	KKM
	Submergence					
10.	AD 22305#	ADT 51/CR 1009 sub-1	152	MS	6335	ADT
11.	CBMAS 22061#	CO 51 / CO 43 sub-1	115	MS	5362	CBE

Checks	:	ADT 51, CR 1009, ADT 52*, CR 1009 sub1
Replications	:	Three
Plot size	:	9 m2
Spacing	:	20 x 15 cm
Locations (8)	:	Aduthurai, Coimbatore, Ambasamudram, Thirupathisaram, Sirugamani, Killikulam, Thanjavur and Needamangalam
Seed despatch	:	5.0 kg to be sent before 20th June

***Will be compared with fine variety ADT 52**

#Trait verification will be done by artificial screening for submergence tolerance

Table 5. MLT- V (Drought - September sowing)

S. No.	Designation	Cross combination	Maturity	Rice grade	Yield (Kg/ha)	Nominating centres
Repeat						
1.	PM 19012	ADT 45 / PM 09001	112	MS	3995	PMK
2.	CB HTR 22011	Anna R 4/Robin HTM	111	MS	5835	CBE
3.	TNAU Pyr 01	IWP/Apo	127	MS	2957	CBE
New						
3.	AD 21098	ADT 43*4/CT 9993	125	MS	5590	ADT
4.	TM 14064	Vandana / IET 23218	115	MS	3590	TKM
5.	CBMAS 22920	IWP / Apo	126	MS	4065	CBE
6.	CBHTR 22012	Anna 4 /Robin HTM	122	SB	5895	CBE
7.	CB 22900	ADT 45*3 /Apo// ADT 45*3/ Wayrarem	112	MS	4609	CBE
8.	PM 18007	ADT 43 / CB 10557	102	MS	4100	PMK
9.	PM 18015	ASD 19 / PMK 3	107	MB	4250	PMK
10.	PM 18026	ADT 43 / CB 10550	105	MB	4333	PMK
Checks	:	IWP, TKM 15 & CO 53				
Replications	:	Three				
Plot size	:	9 m^2 (Direct seeding Natural Environment) 6 m^2 (Rain out Shelter)				
Spacing	:	20 x 10 cm				
Locations (5)	:	Natural condition: Paramakudi, Tirur, Ramnad, Coimbatore & Madurai				
Locations (2)		ROS: Paramakudi and Coimbatore				
Seed despatch	:	4.0 kg to be sent before June 20th				

Table 6. MLT- VI (Bio-fortification)
Both *Kuruvai* (June Sowing) and *Thaladi* (October Sowing)

S. No.	Designation	Cross combination	Maturity	Rice grade	Yield (Kg/ha)	Nominating centres
New						
1.	CB 21102	Paiyur 1 / Kavuni	135	MS	5630	CBE
2.	CB 22105	Pusa Basmati 1 / Kavuni	125	LS	5350	CBE
3.	CBMAS 22051	CB MAS 14065 / Mappillai Samba	136	MS	5999	CBE
4.	TR 21304	CO 51 / IG 39	123	MS	5044	TRY
5.	TR 21318	CO 51 / RPHP 48	126	MS	5101	TRY
6.	AD 21225	IG39 / CO 51	125	MS	5300	ADT
7.	AD 21038	ADT 43/Karppunel	126	MB	5500	ADT
Checks	:	DRR Dhan 45 as micronutrient (Zn) & yield checks CO 57 & ADT 58				
Replications	:	Three				
Plot size	:	9 m^2				
Spacing	:	20 x 10 cm				
Locations (4)	:	Aduthurai, Coimbatore, Tirur, Madurai, Ambasamudram				
Seed despatch	:	3.5 kg to be sent before 20th June				

Table 7. *MLT- VII Special MLT Fast Track (Direct sowing Herbicide trial)

S. No.	Designation	Cross combination	Maturity	Rice Grade	Yield (Kg/ ha)	Nominating centres
	Repeat					
1.	CB HTR 22001	CO 51/Robin HTM	111	MS	5835	CBE
2.	CB HTR 22002	CO 51/Robin HTM	110	MS	4230	CBE
	New					
3.	CB HTR 22022	CBMAS 14110 / HTM	117	MS	5456	CBE

Checks	:	CO 51 and TKM 15
Replications	:	Three
Plot size	:	9 m^2 (Direct seeding Natural Environment)
Spacing	:	20 x 10 cm
Locations (5)	:	Paramakudi, Tirur, Thirupathisaram, Coimbatore and Aduthurai
Seed despatch	:	2.0 kg to be sent before June 20th

*Conducted by Department of Rice

Table 8. @MLT- VIII Drought, Salinity and Submergence (Multiple Abiotic stress tolerant trial)

S. No.	Designation	Cross combination	Maturity	Rice Grade	Yield (Kg/ha)	Nominating centres
	Repeat					
1.	TNAU Pyr 01#*$	CBMAS 14065*3/FL 478 // CBMAS 14065*3 / FR 13A	120	MS	6268	CBE
2.	TNAU Pyr 02#*$	CBMAS 14065*3/FL 478 // CBMAS 14065*3 / FR 13A	120	MS	5697	CBE
3.	TNAU Pyr 03#*$	CBMAS 14065*3/FL 478 // CBMAS 14065*3 / FR 13A	122	MS	5937	CBE
	New					
4.	CBMAS 22061#	CO 51*3 / CO 43 sub-1	115	MS	5362	CBE

Checks	:	CO 51, IWP and CR 1009 Sub 1
Replications	:	Three
Plot size	:	9 m^2
Spacing	:	20 x 10 cm
Locations (7)	:	Submergence: Aduthurai, Needamangalam, Keelvelur Salinity: Trichy, Coimbatore* Drought: Paramakudi and Tirur & Coimbatore (ROS)
Seed despatch	:	2.0 kg to be sent before June 20th

Trait verification for submergence at Aduthurai, in artificial screening facility

* Trait verification for salinity (seedling tray method at Coimbatore)

$ Trait verification for drought at Coimbatore (ROS)

@ Conducted by Dept of Rice, Coimbatore

VII. Rice Multilocation Trials Monitoring Team 2024-25

S. No.	MLT Stations	Monitoring Team
1.	Aduthurai, Thanjavur	Dr. P. Jeyaprakash, Prof (PBG), AD AC & RI Trichy Dr. S. Muthuramu, Assoc. Prof. & Head, CSRC, Ramnad.
2.	Coimbatore	Dr. S. Banumathy, Professor and Head, RRS, Tirur Dr. P. Yogameenakshi, Assoc. Prof. (PBG), RRS, Tirur
3.	Ambasamudram, Killikulam, Thirupathisaram	Dr. R. Suresh, Assoc. Prof. Dept. of Rice, TNAU, CBE Dr. K. Sakthivel, Assoc. Prof. (PB&G), ADAC&RI, Trichy
4.	Madurai and Vaigaidam	Dr. M. Arumugampillai, Prof. (PBG), VOC AC & RI, Killikulam Dr. R. Latha, Assoc. Prof. (PBG), RRS, Thirupathisaram
5.	Paramakudi, Ramanathapuram	Dr. S. Manonmani, P&H, Dept. of Rice, TNAU, CBE Dr. M. Umadevi, AP (PBG), Dept. of Rice, TNAU, CBE
6.	Tirur	Dr. R. Pushpa, AP (PBG), TRRI, Aduthurai Dr. R. Arulmozhi, AP (PBG), TRRI, Aduthurai
7.	Cuddalore	Dr. M. Dhandapani, Assoc. Prof. (PB&G), TRRI, Aduthurai.
8.	Trichy, Sirugamani	Dr. R. P. Gyanamalar, Prof. (PBG), AC & RI, Madurai
9.	Paiyur	Dr. S. Saravanan, Assoc. Prof. & Head, RRS, Ambasamudram

The services of the Entomologist and Pathologist in the MLT conducting centre or nearby research stations may be availed for pest and disease scoring. The Monitoring team will visit at appropriate stage of the trial and report on

1. General Conduct of the Trial a. Plot size and Replications adopted b. Labelling of the plots
2. Admixtures, Disease and Pest Susceptibility if any
3. Top Two entries based on visual observations
4. General Remarks of the trial

B. ACTION PLAN 2024-25

1. THEME I. Development of Climate Resilient Rice Varieties

S. No.	Activity	Action Plan 2023-24
Pyramiding of biotic stress resistance genes		
1.	Development of Biotic stress tolerant rice through pyramiding of genes for BLB, Blast and BPH resistance **Aduthurai:** Dr. M. Dhandapani, ASP (PB&G) Dr. K. Rajappan, Prof. (Plant Pathology) Dr. P. Anandhi, ASP (Agrl. Entomology) **Coimbatore:**	**Aduthurai:** **BLB** The promising from the following crosses • TKM13/AD(Bio)13066//TKM13/PTB33 • TKM13/AD(Bio)13066//TKM13/ADT55 • TKM13/ADT55 // CO52/PTB33 • CO52/PTB33 // CO52/AD(Bio)13066 were evaluated in station trials during *Thaladi*2024. The promising entries tested in IYT during *Thaladi* 2024 will be advanced further. **BPH**

	Dr. R. Suresh, ASP. (PB&G) Dr. C. Gopalakrishnan, Prof. (Pathology) Dr. Sheela Venugopal, AP (Ento.) **Ambasamudhram:** Dr. S. Saravanan, Assoc. Prof. & Head	167 entries were artificial screened and the resistant entries identified are PHS 30 with grade 1.00, PHS 28, PHS 29, PHS 37, PHS 51, PHS 60, PHS 63 and PHS 106 recorded with grade 2.33. Screening and genotyping with gene specific SSR markers will be done to validate the resistance. These resistant entries will be further utilized in crosssing programme. **Coimbatore:** Generation advancement of F_3 families of the cross - APD 19026 / RG 170) / (TKM 13 / AD (Bio) 09518)// (CO52 / PTB 33) to F_4 through marker assisted selection for blast, BLB and BPH. Identified homozygous lines will be forwarded to initial yield trial Sharing of promising materials with Amabasamudhrama and Thirupathisaram
Double trouble (drought + salinity) tolerance		
2.	Development of rice genotypes tolerant to drought and salinity stress through marker aided selection **Coimbatore:** Dr. M. Umadevi, AP (PB&G) Dr. M. Sudha, Asst. Prof. (Pl. Biotech.) **Trichy:** Dr. P. Jayaprakash, Prof. & Head Dept of GPB, ADAC & RI, Trichy **Paramakudi:** Dr. S. Muthuramu, Assoc. Prof. & Head	1. Advancement of Backcross progenies pyramided with QTLs for drought & salinity tolerance. 2. Identification of homozygous lines with drought and salinity and advanced to IYT. 3. Attempting new crossing involving abiotic stress tolerant donors.
DEVELOPMENT OF RICE VARIETIES FOR SUBMERGENCE TOLERANCE		
A.	Evolving breeding lines for submergence tolerance, anaerobic and *in situ* germination **Aduthurai:** Dr. R. Pushpa, AP (PB&G) Dr. R. Arulmozhi, AP (PB&G) **Coimbatore:** Dr. R. Suresh, Assoc. Prof. (PBG) Dr. M. Sudha, Asst. Prof. (Pl. Biotech.)	**Aduthurai** Developing pre-breeding materials using the identified donors. **Action taken** • Evaluation of F_3 population of ADT52/T198 and ADT51/T198 • Introgression of *sub 1* locus into elite long duration-ADT 51, ADT 52. Medium duration-ADT 54 and development of submergence tolerant popular varieties Coimbatore Introgression of *sub 1* locus into medium duration varieties *viz*., CO 52 and ADT 54 and development of submergence tolerant popular varieties

BREAKING THE YIELD BARRIERS IN RICE

S. No.	Activity	Action Plan 2023-24
CREATION OF TNAU 'MAGIC' PLATFORM		
1.	Synthesis of TNAU 'MAGIC' population. **Aduthurai:** Dr. R. Pushpa, AP (PB&G) Dr. R. Arulmozhi, AP (PB&G) **Coimbatore:** Dr. M. Umadevi, AP (PB&G) Dr. R. Suresh, ASP (PB&G) **Tirur:** Dr. S. Banumathy, P&H (PB&G)	**Aduthurai:** Effecting the Four-way crosses and evaluating the four-way crosses of the following crosses: • (AD 16124 X ADT58/ADT 56 x Thandipallian) • (AD 21208 X ADT 58/ AD 16124 x Chirang Sub 1) during *Kharif* 2023 **Tirur:** a.) Effecting the eight-way crosses and evaluating the eight-way crosses of the following crosses: 1. [(IR 20 / TRY3 // CR 1009/Thooyamalli)/ (CB 16533 / Kranthi // NLR 3449 / Norungan)] // [(I.W.Ponni /RNR 15048 // Erramalli/IR 64 Drt)/(CB 14514 / VGD 1 // ADT 37 / Poongar)] 2. [(TKM 13 /CR 1009//ADT 52/Mappilai Samba)/ (I.W.Ponni /RNR 15048 // Erramalli/IR 64 Drt)] // [(I.W.Ponni/RNR 15048 // ADT 37/Poongar)/(Erramalli/IR 64 Drt // CB 12122/WRM 23-25)] b.) Sharing of multiparental cross population to all rice breeding centres of TNAU
UTILIZATION OF TROPICAL JAPONICA LINES IN VARIETAL PROGRAM		
1.	New Generation rice hybrids with high yield potential **Coimbatore:** Dr. S. Manonmani, P & H, Dept. of Rice	Development of three line and two-line rice hybrids utilizing inter sub specific, wild rice and *indica / tropical japo*nica back cross derived restorer / male fertile lines

THEME II Breeding for Market Oriented Rice Varieties (Fine to Super Fine)

S. No.	Activity	Action Plan 2023-24
A.	Development of fine and superfine rice varieties with HRR >60 A%, preferable milling and cooking quality **Aduthurai:** Dr. M. Dhandapani, ASP (PB&G) Dr. R. Pushpa, AP (PB&G) Dr. R. Arulmozhi, AP (PB&G) **Coimbatore:** Dr. R. Suresh, Assoc. Prof (PBG) Dr. M. Umadevi, AP (PB&G) **Tirur:** Dr. P. Yogameenakshi, ASP (PBG) **Killukulam:** Dr. M. Arumugampillai. P&H, (GBP)	**Aduthurai & Coimbatore :** • Effecting new crosses involving elite rice varieties *viz*., CO 52, CO 55, ADT 56, TKM 13, KKM 1 with fine and superfine rice varieties from Andhra Pradesh and Telangana *viz*., BPT 5204, RNR 15048, NDLR7, JGL 1798, Pusa samba 1853 *etc*. • AD18859 : Super fine culture under OFT and ART trials with excellent milling and cooking properties similar to RNR15048 will be promoted through fast track mode for variety release.

THEME III Trait Specific Breeding for DSR

S. No.	Activity	Action Plan 2023-24
A.	Development of rice varieties for DSR with herbicide tolerance, Anaerobic germination, Early vigour **Coimbatore:** Dr. S. Manonmani, P & H Dr. M. Umadevi, AP (PB&G) Dr. R. Suresh, Assoc. Prof. (PBG) Dr. M. Sudha, Asst. Prof. (Pl. Biotech.) **Aduthurai:** Dr. M. Dhandapani, ASP (PB&G) Dr. R. Pushpa, AP (PB&G) Dr. R. Arulmozhi, AP (PB&G) **Paramakudi:** Dr. S. Muthuramu, Assoc. Prof. & Head **Tirur:** Dr. Yogameenakshi, Assoc. Prof. (PBG) **Thrirupathisaram:** Dr. R. Latha, Assoc. Prof. (PBG)	1. Evaluation of cultures introgressed with herbicide tolerant genes under DSR 2. Marker Assisted introgression of genes/QTLs for herbicide tolerance, anaerobic germination and Early vigour into elite backgrounds. 3. Testing of entries in MLT and OFT

THEME IV Development of Therapeutical and Nutritional rich rice varieties

S. No.	Activity	Action Plan 2023-24
A.	Development of bio-fortified rice cultures **Aduthurai:** Dr. R. Pushpa, AP (PB&G) Dr. M. Dhandapani, ASP (PB&G) **Coimbatore:** Dr. M. Umadevi, AP (PB&G) Dr. R. Suresh, Assoc. Prof. (PBG)	Aduthurai A. Conducting MLT – Biofortification B. Evaluation of segregating generation of the following crosses for nutritional traits: ADT 53/Kodai, ADT 54/Kodai, IR 36 / Kodai, TPS 5 / Kodai, ADT 39/ Karuppunellu, ADT 39/Kottanellu, ADT 39/Savulu Samba, ADT 43/Karuppunellu, ADT 43/Kottanellu and ADT 43 / Savulu Samba C. Attempting new crosses involving Fe and Zn rich donors and high yielding parents. **Coimbatore** Attempting new crosses involving high yielding varieties and donors for Fe, Zn and protein *viz*., Mappillai samba, CRR Dhan 310, CRR Dhan 311, Karungkuruvai *etc*.
B.	Release of improved traditional varieties **Aduthurai:** Dr. R. Pushpa, AP, (PB&G) Dr. M. Dhandapani, ASP (PB&G) Dr. R. Arulmozhi, AP (PB&G) **Coimbatore:** Dr. S. Manonmani, P&H, Dept. of	a. Conducting special MLT on PLS b. Improvement of Land races and pure line selection: Aduthuri – Karppukavuni Coimbatore – Mappillai samba Tirur – Manakathai

	Rice Dr. M. Umadevi, AP (PB&G) Dr. R. Suresh, Assoc. Prof. (PBG) **Ambasamudhram** Dr. S. Saravanan, Assoc. Prof. & Head **Tirur:** Dr. S. Banumathy, P&H Dr. Yogameenakshi **Thrirupathisaram:** Dr. Latha, Assoc. Prof (PBG) **Madurai:** Dr. R.P. Gyanamalar, Prof. (PBG) **Killukulam:** Dr. M. Arumugampillai. P&H, (PBG)	Thirupathisaram –Motta kuruva, Boothakalikaruppan Madurai – Chithiraikar

THEME V Development of Pre breeding Genetic Stocks by Exploiting Landraces

S. No.	Activity	Action Plan 2023-24
A.	Exploiting land races for new sources of resistance to Sheath blight, False smut, BPH, Blast & YSB **Coimbatore & Gudalur:** Dr. R. Suresh, Assoc. Prof. (PBG) Dr. M. Umadevi, AP (PB&G) Dr. G. Gopalakrishnan, Prof. (Patho) Dr. Sheelavenugopal, AP (Ento.) **Aduthurai:** Dr. R. Pushpa, AP (PB&G) Dr. R. Arulmozhi, AP (PB&G) Dr. M. Dhandapani, ASP (PB&G) Dr. K. Rajappan, Prof. (Plant Pathology) Dr. P. Anandhi, ASP (Agrl. Entomology) **Ambasamudhram:** Dr. S. Saravanan, Assoc. Prof. & Head Dr. L. Allwin, Assoc. Prof. (Agrl. Ento.) **Tirur:** Dr. Yogameenakshi, Assoc. Prof. (PBG) Dr. V. A. Vijayashanthi, AP (Agrl. Ento.) **Thrirupathisaram:** Dr. Latha, Assoc. Prof. (PBG) Dr. K. Elanchezhyian, Assoc. Prof (Agrl. Ento), AC&RI, KKM	Screening of germplasm lines for Sheath blight, False smut & YSB at hotspot locations Coimbatore Falsesmut: Gudalur, Ambasamudram: Yellow Stem Borer Aduthurai: Blast, Sheath blight, BPH

Accelerated development of rice varieties through Speed breeding

S. No.	Activity	Action Plan 2023-24
A.	Accelerated development of rice varieties **Coimbatore:** Dr. S. Manonmani, P & H Dr. M. Umadevi, AP (PB&G) Dr. R. Suresh, Assoc. Prof. (PBG) Dr. N. Sritharan, Assoc. Prof. (Physiology) Dr. G. SenthilKumar, Assoc. Prof. (Agron)	1. Standardization of RGA Protocol under speed breeding facility 2. Evaluation of segregating populations involving Mappillai Samba

C. Project wise Remarks

S. No.	Project No. & Title	Project Period & PI	Remarks
PROJECTS ON GERMPLASM MAINTENANCE			
ADUTHURAI			
1.	CPBG/ADT/PBG/RIC/2021/001 Characterization of rice germplasm for nutritional properties	June 2021- May 2024 Dr. R. Pushpa, Asst. Prof. (PB&G)	• The project may be closed and completion report may be submitted.
COIMBATORE – CPBG			
1.	CPBG/CBE/RICE/RIC/2021/002 Germplasm collection, evaluation and conservation in rice	June 2021 – May 2024 Dr. R. Suresh Asst. Prof. (PB&G)	• The project may be closed and completion report may be submitted.
PROJECTS ON EVOLUTION			
ADUTHURAI			
1.	CPBG/ADT/PBG/RIC/2022/001: Development of high yielding short duration rice varieties / culture with acceptable grain quality and resistance to major biotic stresses.	June 2022 to May 2027 Dr. R. Pushpa, AP, (PBG) Dr. K. Rajappan, Professor (Pl. Patho.) Dr. P. Anandhi, ASP (Ento.)	• Doner like PTB should be used for crossing. • To be continued
2.	CPBG/ADT/PBG/RIC/2021/001: Development of high yielding, non-lodging, long duration (> 140 days) rice varieties with tolerance to submergence and major biotic stresses	August 2020 to July 2025 Dr. R. Pushpa, Asst. Prof. (PB&G) Dr. P. Anandhi, ASP (Ento.)	• More focus should be given to develop non-lodging, submergence tolerant varieties. • To be continued
3.	CPBG/ADT/RIC/2023/003 Development of medium duration rice varieties suitable for Tamil Nadu	Sep. 2022 to Aug. 2027 Dr. M. Dhandapani, Assoc. Professor (PBG)	• New donors for biotic stresses (BLB, Blast & BPH) may be involved in the crossing programme • To be continued
COIMBATORE – CPBG			
1.	CPBG/CBE/RICE/RIC/2021/006 Developing early maturing (105-	June 2021 – May 2026 Dr. R. Pushpam,	• CO 55 should be popularized

	115 days) rice varieties resistant /tolerant to BPH and blast	Professor (PB&G)	aggressively • To be continued
2.	CPBG/CBE/RICE/RIC/2021/003 Evolution of medium duration rice varieties resistance to blast and false smut	June 2021 – May 2026 Dr. R. Suresh Assoc. Professor (PB&G)	• To be continued
3.	CPBG/CBE/RICE/RIC/2021/004: Development of CMS based rice hybrids with appreciable heterosis for grain yield and acceptable quality	June 2021 to May 2026 Dr. S. Manonmani, Professor (PB&G)	• To be continued
4.	CPBG/CBE/RICE/RIC/2021/005 Development of parental lines for three –line hybrid breeding in rice	June 2021 to May 2026 Dr. S. Manonmani, Professor (PB&G)	• To be continued
5.	CPBG/CBE/RICE/RIC/2022/ 001 Evolution and evaluation of new TGMS lines and two-line TGMS hybrids in rice (URP serial No: URP 2022-0089)	Jan 2022 – Dec 2026 Dr.S.Manonmani, Professor (PB&G) Dr. D. Kumaresan, Professor (PB&G)	• To be continued
COIMBATORE – CPMB&B			
1.	**CPMB/CBE/BIT/RIC/2021/001** Genomic and Structural studies on S-Glutathione transferases to confer herbicide tolerance in rice through computational Approach	2021-23 Dr. N. Saranya Assistant Professor (Bioinformatics)	• To be continued
AMBASAMUTHIRAM			
1.	CPBG/ASD/PBG/Rice/2023/211 Development of elite short duration rice variety bearing desirable yield and cooking quality traits suited for southern districts of Tamil Nadu.	June 2023 - May 2028 Dr. S. Saravanan Associate Professor (PBG) and Head	• Donor parents for major biotic stressed should be used in crossing programme • To be continued
THIRUPATHISARAM			
1.	CPBG/TPS/RIC/2022/001 Evolving medium duration, non-lodging high yielding rice variety suitable for Kumbapoo season in Kanyakumari district	February2022 to January 2025 Dr. R. Latha, Assoc. Professor (PB&G),	• F_1 seeds of CO 47/Boothakalikarup pan may be shared with ADT for inclusion in MAGIC cross • Seeds of Karungkuruvai (Kanyakumari Dt, Type) may be shared with ADT for testing the Fe and Zn content
2.	CPBG/TPS/PBG/RIC/2020/001. Introgression of nonlodging traits in traditional rice varieties Mottakuruva and Kattisamba by mutation breeding	Dec 2020 to Nov 2025 Dr. R. Latha, Assoc. Prof. (PB&G)	•To be continued

Tirur			
1.	CPBG/TKM/PBG/RIC/2020/002 Developing Multi-Parental Advanced Generation Intercross (MAGIC) population in Rice	Dec, 2019 – Nov, 2024 Dr. P. Yogameenakshi, Assoc. Professor (PBG)	• To be continued
2.	CPBG/TIR/PBG/RIC/2021/001 Evolving high yielding medium duration rice varieties with good grain quality suitable for irrigated condition	Sep, 2021-Aug, 2026 Dr. S. Banumathy, Professor (PBG)	• To be continued
3.	CPBG/TIR/PBG/RIC/2023/001 Development of short duration rice varieties with high yield and acceptable grain quality suitable for semidry/irrigated conditions	Dec, 2022 -Nov, 2027 Dr. P. Yogameenakshi, Assoc. Professor (PBG)	• Best cultures for salinity and drought tolerance tested
Madurai			
1.	CPMB/ MDU/ PBG/ RICE / 2023 / 221 Development of high yielding, short duration rice variety suitable for Periyar Vaigai Command Area.	Oct. 2022 – Sept. 2027 Project Leader Dr. R.P. Gnanamalar Professor (PB&G), AC&RI, Madurai Co Project Leader Dr. N. Revathy Prof. (PAT), AC&RI, Madurai	• To be continued
2.	CPBG/MDU/PBG/RIC/2019/001: Evolution of high yielding fine grain quality medium duration rice variety suitable for Periyar Vaigai River Project Area	(Dec. 2018 – Nov. 2023) Dr. A. Sheeba, Asst. Professor (PB&G) Dr. G. Sreenivasan Assoc. Prof. (Agrl. Ento.)	•The project may be closed and completion report may be submitted.
GUDALUR			
1.	CPBG/GDR/PBG/Rice/2020/001. Development and Evaluation of TGMS lines in rice	Sept. 2020 – Aug. 2025 Dr. D. Kumaresan Professor (PBG)	• To be continued
AC & RI, KILLIKULAM			
1.	CPBG/KKM/PBG/RICE/2024/005 Development of high yielding, medium duration rice variety suited to pishanam season in Southern districts of Tamil Nadu	Oct 2023 to Sep 2026 Dr. S. Juliet Hepziba Professor (PB&G)	• To be continued
PARAMAKUDI			
1.	CPBG/PMK/PBG/RIC/2020/002 Development of early duration drought resilient rice genotypes with medium slender grain quality for rainfed rice ecosystem of Ramanathapuram District	Sep. 2020- Aug. 2025 Dr. S. Muthuramu, Assoc. Prof. (PBG) Dr. K. Thangaraj, Professor (PBG),	• To be continued
ADAC & RI,Trichy			
1.	**CPBG/TRY/ PBG/ RIC/2020/002** Development of short duration sodicity tolerant rice varieties (April	Dr. P. Jeyaprakash Professor (PB&G)	• Salinity tolerant entries to be registered with NPBGR and to be

	2020 – March 2025)		given top priority
2.	CPBG/TRY/PBG/ RICE/ 2023/208 Evolution of high yielding medium duration rice genotypes tolerant to sodicity	October 2023 to September 2026 Dr. K. Sakthivel, ASP (PB&G)	• To be continued
SEED CENTRE, TNAU, COIMBATORE			
1.	SEC/CBE/SST/RIC/2022/001 Seed storage studies in newly released Rice varieties of TNAU	Dr. R. Jerlin, Professor (SST), DSST, TNAU, Coimbatore	The project may be continued.
2.	SEC/CBE/SST/RIC/2022/001 Proteomic characterization of selected land races of paddy	Dr. T. Eevera, Assoc. Prof. (SST) DSST, TNAU, Coimbatore	The project may be continued.
MISCELLANEOUS PROJECTS			
ADUTHURAI			
1.	CPBG/ADT/PBG/RIC/2022/New: Nucleus seed production of medium and long duration rice varieties of TRRI, Aduthurai	Aug. 2022 to July 2025 Dr. R. Pushpa Asst. Professor (PBG)	• Project may be continued • More attention and priority should be given to produce genetically pure nucleus seeds
2.	CPBG/ADT/RIC/2023/004: Maintenance breeding of Short Duration Paddy Varieties (TRRI, Aduthurai) through Nucleus Seed production and Supply	July 2022 to May 2027 Dr. M. Dhandapani Asst. Professor (PBG)	
3.	CPBG/ADT/RIC/2023/002: Maintenance breeding of Paddy and Blackgram varieties of TRRI through production and supply of Breeder Seeds	2022- 25 Dr. R. Arulmozhi, Asst. Prof. (PB&G)	
COIMBATORE – CPBG			
1.	CPBG/CBE/RICE/RIC/2021/001 Maintenance breeding of "CO" rice varieties and breeder seed multiplication	June 2021 – May 2024 Dr. S. Manomani Professor (PB&G) Dr. R. Suresh Assoc. Professor (PB&G)	• Project may be continued • More attention and priority should be given to produce genetically pure nucleus seeds
ADAC&RI, TRICHY			
1.	CPBG/TRY/PBG/RIC/2023/001 Nucleus seed production in crop varieties developed at ADAC&RI, Trichy	October 2022 to September 2025 Dr. A. Thanga Hemavathy, ASP (PB&G)	•Project may be continued •More attention and priority should be given to produce genetically pure nucleus seeds
2.	CPBG/CBE/TRY/RIC/2021/001: Breeder seed production and distribution in Rice	June 2021 to May 2024 Dr. A. Thanga Hemavathy, ASP (PB&G)	• Project may be continued
AC&RI, MADURAI			
1.	CPBG/MDU/PBG/ BSP /2020/001 Nucleus and breeder seed production of MDU varieties of rice,	September,2019 to August, 2022 Dr. S. Lakshmi Narayanan	•The project may be closed and should submit completion

	barnyard millet and black gram	Prof. (PB&G)	report
TIRUR			
1.	CPBG/TKM/PBG/RIC/2020/001 Nucleus / Breeder seed production of TKM rice varieties	Dec, 2019 – Nov, 2022 Dr. A. Sheeba, Asst. Professor (PB&G) Dr. S. Banumathy, Professor (PBG)	• The project may be closed and should submit completion report
VAIGAI DAM			
1.	CPBG/VGM/PBG/RIC/2022/001: Breeder Seed Production of Rice VGD 1 and ADT 37	Nov 2021 to Oct 2024 Dr. M. Madhan Mohan, Professor (PBG) & Head	• Project may be continued • More attention and priority should be given to produce genetically pure nucleus seeds
2.	CPBG/VGM/PBG/RIC/2022/002: Breeder Seed Production of paddy varieties ADT (R) 45 and ADT 51	Nov 2021 to Oct 2024 Dr. C. Parameswari, Associate Professor (PBG) Dr. M. Madhan Mohan, Professor (PBG) Dr. M. Jayaramachandran Asst. Professor (PBG)	
AC & RI Killikulam			
1.	CPBG/KKM/PBG/RIC/2017/002 Maintenance breeding for TNAU released rice varieties	Jan 2021- Dec 2026 Dr. S. Saravanan, Assistant Professor (PBG)	• Project may be continued • More attention and priority should be given to produce genetically pure nucleus seeds
PARAMAKUDI			
1.	CPBG/PMK/PBG/RIC/2020/001 Maintenance breeding of rice varieties released from ARS, Paramakudi	Sep, 2020 - Aug, 2025 Dr. S. Muthuramu, Assoc. Professor (PB&G)	• Project may be continued
BHAVANISAGAR			
1.	CPBG/CBE/BSR/RIC/2021/001 Maintenance breeding in popular rice varieties of Tamil Nadu	June 2021 – May 2026 Dr. K. Amudha Assoc. Prof. (PB&G) Dr. S. Utharasu Asst. Prof(PB&G)	• Project may be continued • More attention and priority to be given to produce genetically pure nucleus seeds
PATTUKKOTTAI			
1.	CPBG/PAT/RIC/2023/001-Breeder seed production in Rice	Nov 2022-Oct 2025 Dr. S. Chitra, ASP (PB&G)	• Project may be continued • More attention and priority to be given to produce genetically pure nucleus seeds

II. CROP MANAGEMENT

A. For Adoption/OFT/Information:

FOR ADOPTION:

1. Agronomic efficiency of marine gypsum in sodic soil amelioration in irrigated lowland rice

- Application of marine gypsum @ 4.1 t/ha reduces soil exchangeable sodium percentage from 33.5 to 11.3 by reclamation of sodic soil and 67.4 % grain yield increase over un-reclaimed sodic soil with50% improved agronomic efficiency.
- Marine gypsum can be used as 50% GR (4.1 t/ha) along with Green Manure incorporation @ 6.25 t/ha as alternate source for reclamation of sodic soil.

2. Production of sturdy rice seedlings for machine transplanting

- Fortified seed ($FeSO_4$ @ 0.5% and $ZnSO_4$ @ 0.5%) sowing in nutrient media (*Nutrient media - 70% soil + 20% well decomposed FYM + 10% rice hull + DAP @ 7g + Vermicompost @ 100 g + Azophos @ 14 g per tray of 0.18 m^2) + Vermiwash spray @ 1.0% on 10 DAS in nursery recorded higher seedling vigour Index, planting 3 seedlings/ hill and produced higher growth and yield (5627 kg /ha) and BCR 2.52 of machine transplanted rice

3.Management of grass weed *Leptochloa chinensis* in wet direct seeded rice

- Pre-emergence application of pretilachlor @ 0.75 kg/ha (8 DAS) followed by early post emergence application of Penoxulam + Cyhalofop butyl @ 135 g/ha at 2-3 leaf stage recorded higher weed control efficiency (88%), grain yield (5344 kg/ha), Net return (Rs.68,635/ha) and B:C ratio (2.55)

ON FARM TRIALS (OFT)

OFT 1: Standardization of drip fertigation for short duration paddy variety
Objectives:

- To standardise drip fertigation for short duration paddy (*Kuruvai*/*Navarai*))

Treatments:

T_1: 120:40:40 NPK kg/ha through water soluble fertilizer (Basal 100% of P (SSP)+ N (Urea) & K (SOP) 4 splits: Basal + 3 top dressing through fertigation* once in 6 days

T_2: Straight fertilizers: Blanket recommendation (@ 120: 40:40 kg NPK /ha)

***Fertigation schedule:**

Crop stage	Fertigation Duration (days)	Fertilizer grade
Basal		SSP
Sowing to plant establishment stage (15-27DAS)	18 (3 times)	Urea (21.7 kg/ time)
		SOP (6.6 kg/time)
Active tillering stage (33-45 DAS)	18 (3 times)	Urea (21.7 kg/ time)
		SOP (6.6 kg/time)
Panicle Initiation stage (51-63 DAS)	18 (3 times)	Urea (21.7 kg/ time)
		SOP (6.6 kg/time)
50 % flowering (69 -81 DAS)	18 (3 times)	Urea (21.7 kg/ time)
		SOP (6.6 kg/time)

Observations to be recorded

Growth and yield parameters, Economics, Water indices

Co-ordinating centre & Scientist in-charge:	CSRC, Ramanathapuram	Dr. S. Vallal Kannan, Associate Professor (Agronomy)
Centres & Scientist in-charge:	KVK, Madurai	Dr. E. Subramanian, Programme Coordinator
	AC&RI, Killikulam	Dr. M. Joseph, Prof. (Agronomy)

OFT 2: Drone application of liquid formulation of rice booster for yield enhancement in rice

Objectives:

To assess the impact of liquid formulation of rice booster through drone application on growth and yield of rice

Treatments:

T_1: 1 % Rice bloom by hand spray

T_2: 4 % liquid formulation of rice booster by drone spray

Stages of Spray:

- Flower initiation [Heading] and grain filling stage

Observations to be recorded:

- Yield parameters
- Economics

Co-ordinating centre & Scientist in-charge:	Department of Crop Physiology, TNAU, Cbe	Dr. V. Ravichandran, Prof. (CRP)
Centres & Scientist in-charge:	TRRI, Aduthurai	Dr. S. Elamathi, Assoc. Prof. (Agronomy)
	KVK, Needamangalam	Dr. V. Karunakaran, Assistant Professor (Agronomy)
	ARS, Kattuthotam	Dr. T. Parthipan, Assistant Professor (Agronomy)
	IoA, Kumulur	Dr. S. D. Sivakumar,

		Prof. (Agron.) & Principal
	ARS, Vaigaidam	Dr. G. Sudhakar, Associate Professor (Agronomy)
	KVK, Thirupathisaram	Dr. A. Selvarani, Assistant Professor (Agronomy)

FOR INFORMATION

1. Agronomic management of yellowing symptom in rabi rice

Application of 100% RDF + $ZnSO_4$ @ 25 kg/ha + 1% Rice foliar formulation (MAP 0.5 %+ 0.25 % KCl + 0.25 % $MgSO_4$) recorded higher grain yield and it was on par with STCR-IPNS based NPK application + foliar spray of 0.25 % $CuSO_4$ + 0.5 % $ZnSO_4$.

2. Innovative labour-saving technique in Direct seeded Rice

Direct seeding of 40 kg/ha using drone recorded higher grain yield (5017 kg/ha), Net return (Rs.60,985) and BCR (2.45)

- Labour saving for sowing (2-man days) compared with paddy drum sowing
- Time saving for sowing (7 h/ha)

3. Soil weed seed bank, weed dynamics, pests load, soil fertility and productivity as influenced by rice-duck farming

Rice + Duck farming (100 ducklings /ha) for 3 hours at 4 days interval for 20 days (20 DAT to PI stage) resulted in

- Reduced weed seed bank (55 % reduction of *Echinochloa colona* and 84 % reduction of *Cyperus difformis*)
- Higher weed control efficiency of 57.7%
- Less incidence of pests (stem borer (1.2 %) and leaf folder (0.45 %))
- Increased soil nutrient status (389 kg ha^{-1} N and K 540 kg ha^{-1}K)
- Higher grain yield (6333 kg/ha) with BCR (2.26)

4. Evaluation of Imazethapyr herbicide tolerant genotype under direct seeded rice ecosystem

- Culture CB HTR 22002 and CB HTR 22001 showed no phyto-toxicity to Imazethapyr spray which resulted higher WCE (98.8 and 98.7%) and Grain yield (4285 and 4026 kg ha^{-1}).

5. Performance of Improved *Kavuni* (CO 57) under Organic Cultivation

- CO 57 is identified as highly suitable variety for organic cultivation and recorded higher grain yield of 4378 kg ha^{-1} and higher net returns Rs. 1,19,747 /- ha with the B:C ratio of 2.23.

B. Action Plan (2024-2025)

Action Plan 1: Economic evaluation of foliar Super Nano N&K fertilizer through drone in Wet direct seeded rice

Objectives:

- To study of the effect of drone applied Nano fertilizer at on growth and yield of Wet DSR
- To find out the effect Nano source of N&K on nutrient use efficiency in Wet DSR
- To workout the economics of various treatments

Treatment details

T_1	100% NPK soil application
T_2	75% N&K soil application + Foliar spray of Nano N&K @ 2 times (AT & 15 DA first spray)
T_3	75 % N&K soil application + Foliar spray of Nano N&K @ 3 times (AT & 15 days intervals thereafter)
T_4	50% N&K soil application + foliar spray of Nano N&K @ 2 times (AT & 15 DA first spray)
T_5	50% N&K soil application + foliar spray of Nano N&K @ 3 times (AT & 15 days intervals thereafter)
T_6	Foliar spray of Nano fertilizer NPK (Nano Urea, Nano DAP & Nano K @ 3 times (AT & 15 days intervals thereafter)
T_7	Absolute control

Observations to be recorded:

Soil analysis: pH, EC and Available N, P, K (pre and post soil)
Use Efficiencies: Nutrient use efficiency (N&K) and Agronomic use efficiency
Crop parameters (AT, PI and at harvest): Plant height (cm); Dry matter production (DMP); Number of tillers m^{-2}, Root length & volume, Yield attributes and yield Economics

Co-ordinating centre & Scientist In-charge:	AC&RI, Killikulam	Dr. M. Joseph, Professor (Agronomy)
Centres & Scientist In-charge:	TRRI, Aduthurai	Dr. S. Elamathi, Assoc. Prof. (Agron.)
	Dept. of Agronomy, TNAU, Coimbatore	Dr. G. Senthil Kumar, Assoc. Prof. (Agron.)
	KVK, Madurai	Dr. E. Subramanian, Programme Coordinator
	ADAC&RI, Trichy	Dr. T. Ramesh, Assoc. Prof. (AGR),
	ARS, Thanjavur	Dr. T. Parthipan, AP (Agronomy)

Action Plan 2: Land management and seed treatment techniques in dry direct seeded rice (Seed drill sowing)

Preamble:

- Poor seed germinations and uneven crop stand are main constraints in DSR
- In Dry DSR, Firm and well pulverized seed bed is necessary for adequate seed to soil contact for achieving good crop stand
- Land Levelling is considered a pre-requisite for mechanized sowing
- Seed Priming techniques used to reduce germination time, uniform germination and better seedling vigour

Objective:

- To find out the effect of laser levelling and seed hardening techniques on growth and yield of dry DSR.

Treatment details:

A. Land Management:

M_1: Laser Levelling

M_2: Conventional Method

B. Seed treatment/ Seed hardening *

S_1: Vidhai Amirtham 25g/kg of seed

S_2: 1% KCl

S_3: 1 % PPFM

S_4: Control

***Method of seed treatment**

Vidhai Amirtham 25g/kg of seed: Add required quantity of TNAU Vidhai Amirtham (25g / Kg of seed) to the seeds and stir vigorously for complete coating of seeds. After drying, the coated seeds should be used for sowing

1% KCl: Seeds should be soaked for 12 hours and dried back to original moisture content. Dried seeds should be used for sowing

1 % PPFM: Imbibe seeds in 1% volume for 10 minutes and dried back to original moisture content. Dried seeds should be used for sowing

Observation to be recorded

i. Germination percentage %
ii. Seedling emergence rate
iii. Seedling vigour
iv. Growth and yield parameter

Design: Strip Plot, **Replication:** 4

Duration: 2024-26, **Season:** *Kuruvai* or *Kharif*

Co-ordinating centre & Scientist In-charge:	TRRI, Aduthurai	Dr. S. Elamathi, Associate Professor (Agronomy) Dr. G. Sathiyanarayanan, Associate Professor (SST)

Centres & Scientist In-charge:	KVK, Madurai	Dr. E. Subramanian, Programme Coordinator
	ARS, Vaigai Dam	Dr. M.P. Kavitha, Assoc. Prof. (Agron.)
	AC&RI, Killikulam	Dr. M. Hemalatha, Prof. (Agron.)
	ARS, Thanjavur	Dr. T. Parthipan, Asst. Prof. (Agron.)

Action Plan 3: Efficacy of newer herbicide molecules in machine sown in dry direct seeded rice

Preamble:

- Direct seeded rice encounters complex weed flora and controlling weeds is highly essential.
- Herbicide applied alone or in combination as pre-emergence or post emergence or sequential application of newer herbicide molecules are highly warranted to get efficiency and yield

Objective:

- To find out the effect of weed management practices on weed dynamics and yield under dry direct seeded rice system Treatment details

Treatment details:

T_1:	Pendimethalin @ 0.75kg /ha (PE) alone
T_2:	Triafamone 20% + Ethoxysulfuron 10% WG @ 67.5 g/ha (EPoE)
T_3:	Florpyrauxifen-benzyl (2.13% w/w) + Cyhalofop-butyl (10.64 % w/w) EC @ 20.5 g /ha (EPoE)
T_4:	Pendimethalin 0.75kg ai/ha (PE) *fb*Triafamone 20% + Ethoxysulfuron 10% WG @ 67.5 g/ha (EPoE)
T_5:	Pendimethalin 0.75kg ai/ha (PE) *fb*Florpyrauxifen-benzyl (2.13% w/w) + Cyhalofop-butyl (10.64 % w/w) EC @ 20.5 g ai/ha (EPoE)
T_6:	Pendimethalin @ 0.75kg /ha (PE) *fb* Cono weeding at 25 DAS
T_7:	Weedy check
T_8:	Weed free

Observation to be recorded

i. Weed Flora
ii. Weed Dry weight
iii. Weed index
iv. Weed control efficiency
v. Phytotoxicity
vi. Growth and yield parameters

Design: RBD

Replications: Three replications

Duration: 2024-26,
Season: *Kuruvai or Kharif*

Co-ordinating centre & Scientist in-charge:	TRRI, Aduthurai	Dr. S. Elamathi, Associate Professor (Agronomy)
Centres & Scientist In-charge:	Dept. of Agronomy, TNAU, Coimbatore	Dr. G. Senthil Kumar, Assoc. Professor (Agronomy)
	ARS, Vaigai Dam	Dr. G. Sudhagar, Assoc. Professor (Agronomy)
	AC&RI, Madurai	Dr. T. Sampath Kumar, Assistant Professor (Agronomy)
	AC&RI, Killikulam	Dr. M. Hemalatha, Prof. (Agron.)
	KVK, Needamangalam	Dr. V. Karunakaran, Assistant Professor (Agronomy)
	ARS, Thanjavur	Dr. T. Parthipan, Assistant Professor (Agronomy)

Action Plan 4: Comparative evaluation of organic and natural farming practices in rice

Preamble

As natural farming is gaining momentum, the natural farming package of practices developed by NOFRC will be evaluated and compared with the existing organic package of practices in terms of plant growth, yield and soil health parameters.

Objective

To evaluate and validate the natural farming package of practices in rice.

Treatment details

- T_1 - Organic package of practices
- T_2 - Natural farming package of practices
- T_3 - Control

Duration: One year (2024 - 2025)

Variety: CO 57 (Improved *Kavuni*)

Package of practices for organic rice cultivation

- Seed treatment (g/kg): *Bacillus subtilis* 10 g; Azospirillum 30 g; Phosphobacteria 30 g
- Green manure crop: *In-situ* incorporation of Sesbania aculeate
- Basal application: Rock phosphate @ 250 kg / ha; Neem cake @ 250 kg / ha
- Soil application: Azospirillum @ 2.5 kg / ha; Phosphobacteria @ 2.5 kg / ha; Split application of vermicompost @ 1 t / ha each at maximum tillering & panicle initiation stage
- Growth promotion: Panchagavya @ 3 % at 30 and 45 DAT as foliar spray
- Insect pest management: NSKE @ 5 % / Neem oil @ 3 %
- Disease management: *Bacillus subtilis* (0.2 %)

Package of practices for rice cultivation under natural farming

- Seed and seedling treatment with beejamrit @ 5 litres per 25 kg of seed
- Application of ghanjeevamrit @ 1000-1500 kg/acre during last ploughing or puddling.

- Soil application of jeevamrit @ 600 litres/acre along with irrigation water, four times, at 35, 50, 65 and 80 days after transplanting (DAT).
- Foliar application of 50 litres of jeevamrit in 100 litres of water four times at 25, 45, 55, 70 days after transplanting.
- Application of azolla at 10 kg/acre after 7 DAT which fixes nitrogen, reduces weed development, acts as live organic mulch (reduces irrigation frequency by reducing evaporation loss of water) and the biomass can be incorporated.
- Non-negotiable practices like clipping of leaf tips, alleys, border / bund / peripheral plantation with marigold, yellow sticky traps, pheromone traps for yellow stem borer, bird perches and light traps must be practiced.
- Application of panchagavya @ 3 litres/acre, twice at tillering stage and 70 DAT, egg amino acid or fish amino acid @ 2 litres/acre, once at panicle initiation stage and saptdhadhanyakura tonic @ 700 grams of paste in 100 litres of water, once at milking and grain filling stage to boost both quality and quantity of rice.
- Insect pest management: Agniastra / brahmastra @ 3 l/ac or neem astra @ 200 l/acre
- Disease management: Sounthaster (dry ginger - milk extract) @ 6 l/acre

Observations to be recorded

Growth characters: Plant height at harvest, DMP at harvest

Yield characters: Days to 50% flowering, No. of productive tillers / hill, No. of grains & filled grains / panicle, test weight, grain yield, straw yield and HI

Economics: Cost of cultivation, Net returns with and without premium price (25%) and BCR

Centre & Scientist in-charge:	TNAU, Coimbatore	Dr. R. Krishnan, Professor and Head, NOFRC Dr. M. Suganthy, Professor (Agrl. Entomology), NOFRC Dr. S. Manonmani, Professor and Head, Department of Rice Dr. G. Senthil Kumar, Assoc. Prof. (Agron.), Dept. of Rice

Action Plan 5: Impact of nutrient microbial consortia for enhancement of seedling vigour under direct seeded rice

Objective:

- To study the impact of nutrient-microbial consortia for enhancing early seedling vigour and plant growth in direct seeded rice.

Treatment details:

T_1	:	Control
T_2	:	Seed treatment with Formulation I
T_3	:	Foliar spray of Formulation II at 21 DAS

Observations to be recorded:

- Germination percentage
- Seedling vigour
- Chlorophyll content in seedling stage

Duration: 2024-26

Co-ordinating centre & Scientists in-charge:	Dept. of Rice, TNAU, Coimbatore	Dr. N. Sritharan, Assoc. Prof. (Crop Physiology) Dr. G. Senthil Kumar, Assoc. Prof. (Agronomy)
	AC&RI, Madurai	Dr. K.G. Sabarinathan, Assoc. Prof. and Head (Agrl. Microbiology)

C. RESEARCH PROJECTS AND REMARKS				
I. ACTION PLAN				
S. No.	**Title**	**Scientist involved / Lead scientist**	**Duration**	**Remarks**
1.	Agronomic management of yellowing symptom in rabi rice	**Co-ordinating centre & Scientist in-charge** Dr. M. Joseph, Professor (Agronomy), AC&RI, Killikulam Dr. S. Srinivasan, Prof., Dept. of Crop Physiology and Biochemistry, AC&RI, Killikulam **Sub-centre & Scientist In-charge** Dr. S. Jothimani, Prof. & Head, Dept. of SS&AC, AC&RI, Killikulam Dr. M. Paramasivan, Assistant Prof. Dept. of SS&AC, AC&RI, Killikulam	2022-2024	• To be closed and given for Information
2.	Innovative labour saving technique in Direct seeded Rice	**Co-ordinating centre & Scientist In-charge** Dr. S. Radhamani, Professor (Agron), Dept of Agronomy, TNAU, Coimbatore Dr. G. Senthil Kumar, Assoc. Prof. (Agron.), Dept of Rice*, TNAU, Coimbatore Dr. R. Kumaraperumal, Assoc. Prof. (SS&AC), Dept of RS & GIS, TNAU, Coimbatore Dr. P. Murali Arthanari, Prof. (AGR), Dept of Agronomy, AC & RI, Coimbatore **Centres & Scientist In-charge** Dr. S. Elamathi, Assoc. Prof. (AGR), TRRI, Aduthurai Dr. T. Ramesh, Assoc. Prof. (AGR), ADAC&RI, Trichy Dr. M. Hemalatha, Prof. (Agronomy), AC&RI, Killikulam	2022-2024	• To be closed and given for Information
3.	Soil weed seed bank, weed dynamics, pests load, soil fertility and productivity as influenced by rice-duck farming	**Co-ordinating centre & Scientist In-charge** Dr. G. Prabu kumar, Associate Professor (Agronomy), Department of Agronomy, CFU, TNAU, Coimbatore	2022-2024	• To be closed and given for Information
4.	Standardization of drip fertigation for short duration paddy	**Co-ordinating centre & Scientist In-charge** Dr. S. Vallal Kannan, Programme Co-ordinator, KVK, Ramanathapuram	2022-2024	• Recommended for OFT

5.	Management of grass weed *Leptochloa chinensis* in wet direct seeded rice	**Co-ordinating centre & Scientist In-charge** Dr. S. Radhamani, Professor (Agron), Dept of Agronomy, TNAU, Coimbatore Dr. G. Senthil Kumar, Assoc. Professor (Agron.), Dept of Agronomy, TNAU, Coimbatore **Sub-Centres & Scientist In-charge** Dr. R. Nageswari, Associate Professor (Agronomy), TRRI, Aduthurai Dr. M. Joseph, Professor (Agronomy), AC&RI, Killikulam Dr. K. Sivagami, AP (Agronomy), RRS, Tirur Dr. S. Vallal Kannan, Programme Co-ordinator, KVK, Ramnad	2023-2024	• Recommended for Adoption
6.	Evaluation of Imazethapyr herbicide tolerant genotype under direct seeded rice ecosystem	**Co-ordinating centre & Scientist In-charge** Dr. G. Senthil Kumar, Assoc. Professor (Agron.), Dept of Agronomy, TNAU, Coimbatore **Sub-centres & Scientists In-charge** Dr. S. Elamathi, Assoc. Prof. (Agron.), TRRI, Aduthurai Dr. A. Alagesan, Assoc. Prof. (Agron.), AC&RI, Nagapattinam Dr. K. Sivagami, Asst. Prof. (Agron.), KVK, Tirur Dr. S. Vallal Kannan, Prog. Co-ordinator, KVK, Ramnad	2023-2024	• To be closed and given for Information
7.	Performance of Improved *Kavuni* (CO 57) under Organic Cultivation	**Co-ordinating centre & Scientist In-charge** NOFRC, TNAU, Coimbatore Dr. R. Krishnan, Professor and Head Dr. M. Suganthy, Professor (Agrl. Entomology) Dr. G. Senthil Kumar, Assoc. Prof. (Agronomy) Dr. R. Nageswari, Assoc. Prof. (Agronomy) Dr. J. Bhuvaneswari, Assistant Professor (Agronomy) Dr. N. Sakthivel, Professor (Agronomy) Dr. K. Sivagamy, Assistant Professor (Agronomy)	2023-2024	• To be closed and given for Information
1.	Drone application of liquid formulation of rice booster for yield enhancement in rice	**Co-ordinating centre & Scientist In-charge** **Department of Crop Physiology, TNAU, Coimbatore** Dr. V. Ravichandran, Prof. (CRP)	2023-2024	• Recommended for OFT

		Dr. D. Vijayalakshmi, Prof. (CRP) **Centres & Scientists In-charge** **RRS, Aruppukottai** Dr. K. Krishnasurendar, Asst. Prof. (CRP) **KVK, Sirugamani** Dr. S. Nithila, Associate Professor, (CRP) **AC& RI, Eachangkottai** Dr. C. Tamilselvi, Asst. Prof. (CRP)		

II. ON FARM TRIAL				
1.	Evaluating the agronomic efficiency of marine gypsum in sodic soil amelioration in irrigated lowland rice	**Co-ordinating centre & Scientist in-charge** **ADAC&RI, Trichy** Dr. S. Rathika, Assoc. Prof. (AGR) Dr. M. Baskar, Prof. & Head, Dept. of SS&AC **Centres & Scientist in-charge** **AC&RI, Nagapattinam** Dr. A. Alagesan, Assoc. Prof. (AGR) Dr. A. Anuratha, Assoc. Prof. (SS&AC) **CSRC/KVK, Ramanathapuram** Dr. S. Vallalkannan, Programme Co-ordinator Dr. T. Balaji, Asst. Prof. (SS&AC)	2023-2024	• Recommended for Adoption
2.	Production of sturdy rice seedlings for machine transplanting	**Co-ordinating centre & Scientist in-charge** **Department of Agronomy, VOC AC&RI, Killikulam** Dr. M. Hemalatha, Prof. & Head (Agron.), VOC AC&RI, KKM Dr. G. Senthil Kumar, Assoc. Prof. (Agron.), Dept. of Rice, TNAU, Coimbatore Dr. T. Ramesh, Assoc. Prof. (Agron.) & Head, Dept. of Agronomy, ADAC&RI, Trichy Dr. K. Sivagamy, Asst. Prof. (Agron.), KVK, Tirur. Dr. T. Sampathkumar, Asst. Prof. (Agron.), AC&RI, MDU, Dr. P. Rajarathinam, Assoc. Prof. (Agron.), TRRI, ADT	2023-2024	• Recommended for Adoption

III. UNIVERSITY RESEARCH PROJECTS				
1.	DCM/KUM/RIC/2021/001 - Studies on the response of traditional rice cultivars under organic production systems	**Project Leader** Dr. S.D. Sivakumar, Prof. (Agron.) & Principal, IOA, Kumulur **Co-Project Leaders** Dr. R. Vinoth, Teaching Assistant (PBG) Dr. A. Vijayaprabhakar, TA (Agron.)	2021-2024	• To be closed and given for Information
2.	DCM/ADT/AGR/RICE/2023/137: Assessment of crop duration of promising rice varieties grown in *Kuruvai* season and Samba season in Cauvery Delta Zone (CDZ)	**Project Leader** Dr. R. Nageswari, Associate Professor (Agron), TRRI, Aduthurai **Co-Project Leaders** Dr. R. Arulmozhi, Asst. Prof. TRRI, Aduthurai Dr. C. Tamil Selvi, Assistant Professor (Crop Physiology), ICAR-Krishi Vigyan Kendra, Tirur.	2023-2025	• To be Continued
3.	DCM/TRY/AGR/RICE/2023/239 Studies on the performance of traditional rice landraces for sodicity tolerance	**Project Leaders** Dr. S. Rathika, Assoc. Professor (Agronomy) **Co-Project Leaders** Dr. M. Baskar, Professor & Head (SS&AC) Dr. V. Dhanushkodi, Assoc. Prof. (SS&AC)	2023-2025	• To be Continued
4.	DCM/CBE/CRP/RICE/2023/138 - Prevention of viviparous germination in rice by foliar sprays of chemicals and growth retardants	**Project Leader** Dr. D. Vijayalakshmi, Prof. (CRP), Dept. of Crop Physiology, TNAU, Coimbatore.	2023-2025	• To be Continued
5.	DCM/TIR/AGR/Rice/2023/188 - Evaluation of Weed Management practices in Direct sown rice of Tiruvallur District	**Project Leader** Dr. K. Sivagamy, Assistant Professor (Agronomy), KVK, Tirur	2023-2024	• To be Continued
6.	DCM/TRY/AGRO/RIC/2023/001- Effect of nano urea on the yield of rice in sodic soil	**Project Leader** Dr. T. Ramesh, Assoc. Prof. and Head, Dept. of Agronomy, ADAC&RI, Tiruchirappalli	2023-2025	• To be closed and completion report may be submitted
7.	DCM/MDU/AGR/RICE/2023/156. Standardization of crop geometry, time of Brown and Green manuring of daincha in aerobic rice.	**Project Leader** Dr. E. Subramanian, Associate Professor (Agronomy), AC&RI, Madurai	2023-2025	• To be continued • Biomass and nutrient addition to be calculated • Unweeded check

				may be included to work out the WCE
8.	DCM/ ADT/ AGR/ RICE/ 2023/ 180 - Integrated weed management of water lettuce (*Pistia stratiotes*) in direct sown wet seeded rice.	**Project Leader** Dr. P. Rajarathinam, Associate Professor (Agronomy), TRRI, Aduthurai Dr. T. Sivasankari Devi, Asst. Professor (Agrl. Microbiology), TRRI, Aduthurai	2023-2026	• To be Continued
IV. ALL INDIA COORDINATED RESEARCH PROJECTS				
1.	AICRP /PBG /ADT / RIC/ 002 - Nutrient response trials on selected AVT 2 rice cultures under high and low input management IME	Dr. S. Elamathi, Associate Professor (Agronomy), TRRI, Aduthurai	2024-2025	• To be Continued
2.	AICRP /PBG /ADT / RIC/ 002 - Nutrient response trials on selected AVT 2 rice cultures under high and low input management LATE	Dr. S. Elamathi, Associate Professor (Agronomy), TRRI, Aduthurai	2024-2025	• To be Continued
3.	AICRP /PBG /ADT / RIC/ 002 - Developing a suitable package of practices for higher yield in wet DSR system	Dr. S. Elamathi, Associate Professor (Agronomy), TRRI, Aduthurai	2024-2025	• To be Continued
4.	AICRP /PBG /ADT / RIC/ 002 - Long term trial on weed dynamics in mono or double cropped rice system under different establishment methods	Dr. S. Elamathi, Associate Professor (Agronomy), TRRI, Aduthurai	2024-2025	• To be Continued
5.	AICRP /PBG /ADT / RIC/ 002 Water management for enhancing water use efficiency and productivity of mechanical transplanted rice	Dr. S. Elamathi, Associate Professor (Agronomy), Tamil Nadu Rice Research Institute, Aduthurai	2024-2025	• To be Continued
6.	AICRP /PBG /CBE / RIC/ 003 - Nutrient Management Trials – AVT 2 – Early (TP) - Nutrient response trials on selected AVT 2 rice cultures under optimum and low input management	Dr. G. Senthil Kumar, Associate Professor (Agronomy), Department of Rice, TNAU, Coimbatore.	2024-2025	• To be Continued

7.	AICRP /PBG /CBE / RIC/ 003 - Nutrient Management Trials – AVT 2 – Irrigated (Medium)Nutrient response trials on selected AVT 2 rice cultures under optimum and low input management	Dr. G. Senthil Kumar, Associate Professor (Agronomy), Department of Rice, TNAU, Coimbatore.	2024-2025	• To be Continued
8.	AICRP /PBG /CBE / RIC/ 003 - Nano fertilizers for increasing nutrient use efficiency, yield and economic returns in transplanted rice	Dr. G. Senthil Kumar, Associate Professor (Agronomy), Department of Rice, TNAU, Coimbatore.	2024-2025	• To be Continued
9.	AICRP /PBG /CBE / RIC/ 003 - Developing a suitable package of practices for higher yield in wet direct seeded rice (DSR) system	Dr. G. Senthil Kumar, Associate Professor (Agronomy), Department of Rice, TNAU, Coimbatore	2024-2025	• To be Continued
10.	AICRP/PBG/CBE/RIC/003 - Influence of silicon on improving abiotic stress tolerance in rice genotypes	Dr. N. Sritharan, Associate Professor (Crop Physiology), Department of Rice, CPBG, TNAU, Coimbatore	2024-2025	• To be Continued
11.	AICRP/PBG/CBE/RIC/003 - Screening for submergence tolerance in rice genotypes	Dr. N. Sritharan, Assoc. Prof. (Crop Physiology), Dept. of Rice, CPBG, TNAU, Cbe	2024-2025	• To be Continued
12.	AICRP/PBG/CBE/RIC/003 - Physiological characterization of selected rice genotypes for multiple abiotic stress tolerance	Dr. N. Sritharan, Assoc. Prof. (Crop Physiology), Dept. of Rice, CPBG, TNAU, Cbe	2024-2025	• To be Continued
V. EXTERNALLY FUNDED PROJECTS				
1.	TNSLURB/ACRI/KKM/2021/R001- Water budgeting and economical assessment of irrigation water through drip fertigation in Rice – Rice – Pulse cropping system	**Co-ordinating centre & Scientist in-charge** **PI:** Dr. M. Joseph, Professor (Agronomy), AC&RI, Madurai **Co-PI:** Dr. S. Srinivasan, Prof. & Head (Crop Physiology & Biochem.), AC&RI, Killikulam	2021-2023	• To be continued
2.	Assessing the water productivity, soil health, yield and economics of rice – based cropping systems under transplanted	**Co-ordinating centre & Scientist in-charge** Dr. K. Subrahmaniyan, Director, TRRI, Aduthurai	2023-2025	• To be continued

III. NATURAL RESOURCE MANAGEMENT

DEPARTMENT OF SOIL SCIENCE AND AGRICULTURAL CHEMISTRY

A1. For Adoption

1. Skipping of Phosphorus on Rice Productivity in High P status soil

STCR-IPNS recommendation recorded the highest grain yield of 5862 kg ha^{-1} followed by Blanket recommendation along with FYM @ 12.5 kg ha^{-1} (5612 kg ha^{-1}). STCR recommendation with skipping of 10 kg ha^{-1} of P in high P status soil could sustain the grain yield. The highest BCR (2.50) was recorded with STCR recommendation with skipping of 10 kg ha^{-1} of P.

2. Zinc nutrition for direct sown Rice through Zinc lysinate

Zinc lysinate, a chelated form of Zinc containing 20% Zinc. The application of recommended fertilizers (RDF) combined with Zinc lysinate at 2.5 kg ha^{-1}along with foliar spray @ 0.2% Zinc lysinate twice at AT and PI stages recorded the highest grain yield of 4352 kg ha^{-1}, a Zinc content of 14.3 ppm, and total Zinc uptake of 104 g ha^{-1} in direct sown Rice with a BCR of 2.27. Application of RDF + Zinc lysinate @ 2.5 kg ha^{-1} recorded 10.6 % higher grain yield than RDF + $ZnSO_4$@ 25 kg ha^{-1}.

A2. OFT for the year 2024-2025

1. Assessment of Nutrient Use Efficiency of TNAU – WSF in Direct Sown Rice under Drip Irrigation

Objectives

- To assess the effect of TNAU – WSF on growth and yield of rice
- To study the effect of TNAU- WSF on nutrient use efficiency and soil properties

Project period: 2024 – 2025

Treatment details

T_1 : Flooded irrigation + RDF

T_2 : Drip irrigation + RDF

T_3 : 50 kg ha^{-1} TNAU- WSF through drip fertigation and remaining N by urea through drip

Lead centre and Scientists	AC&RI, Madurai	Dr. S. Sheeba, Professor (SS&AC) Dr. J. Prabhaharan, Assoc. Prof. (SS&AC)
Co-ordinating centre & Scientists in-charge:	AC&RI, Killikulam	Dr. K. Manikandan, AP (SS&AC)

A3. For Information

1. Effect of Fly Ash on Improving Soil Properties and Rice Productivity

Application of Fly Ash @ 15 t ha^{-1} + FYM @ 12.5 t ha^{-1} + SSB recorded the highest grain (7330 kg ha^{-1}) and straw yield (8782 kg ha^{-1}) of rice with a yield increase of 49 % over absolute control with BCR (2.03) followed by Fly Ash @ 20t ha^{-1} + FYM@ 12.5 t ha^{-1} + SSB. The control plot treatment was found to be moderately susceptible and Fly Ash treated plot was moderately resistant to ear head bug incidence, whereas the control plot was moderately resistant and Fly Ash treated plot was resistant to stem borer.

2. Evaluating the Potential Rice Varieties Suitable for Saline and Sodic soil condition

The rice varieties CSR 46 and TRY 3 were tolerant up to soil EC of 6 $dSm^{-1}$and, CO 56 and ADT 58 up to soil EC 2 dS $m^{-1}$and could achieve 50 per cent yield. Under sodic condition, the rice variety TRY 3 was tolerant up to the ESP of 32 and CO 56, CSR 46 and ADT 58 up to ESP of 24to achieve 50 per cent yield

3. Management of Multi-nutrient disorder in Rice

Application of STCR-IPNS based fertilizer along with foliar application of 0.25 % $CuSO_4$ + 0.5 % $ZnSO_4$ registered higher grain yield and soil nutrient status in Aduthurai and Killikulam. Whereas at Trichy, STCR based NPK + foliar 1% $MgSO_4$ + 0.25 % $CuSO_4$ + 0.5 % $ZnSO_4$ produced higher grain yield and soil nutrient status.

4. Assessment of NUE of TNAU – WSF in Direct sown Rice under Drip irrigation

Application of TNAU WSF @ 50 kg ha^{-1} through drip fertigation (remaining N – fertigation through urea) recorded the highest net income (Rs. 53,515 ha^{-1}), BCR (2.38), WUE (7.49 kg ha^{-1} mm), WP (113.4 kg ha^{-1} mm) along with 15 % water saving in rice. TNAU WSF @ 40 and 50 kg ha^{-1} recorded an additional income of Rs.8, 331/- and Rs.12,249/- respectively when compared to 100 per cent conventional fertilisers through surface irrigation. Application of 50 kg ha^{-1} TNAU WSF resulted in saving of 130 kg of urea when compared to the conventional fertilisers.

5. Permanent Manurial Experiment in Rice based Cropping system

In Aduthurai, the results of 32 years old continuous rice cropping (64 crops) (PME experiment) revealed that higher yields of 6279 kg ha^{-1} were achieved in *Kuruvai* with 125:50:50 kg NPK+Green manure @ 6.25 t ha^{-1}+Gypsum @ 500 kg ha^{-1} and in *Thaladi* (6163 kg ha^{-1}) with 150:60:60 kg NPK+FYM @ 12.5 t ha^{-1}+ Gypsum @ 500 kg ha^{-1}. The soil organic carbon increased to 10.8 g kg^{-1} from an initial value of 5.7 g kg^{-1}, and available nitrogen and phosphorus registered a positive balance under Integrated Nutrient Management (INM). Phosphorus application may be skipped in consecutive cropping where soil available phosphorus is high.

6. Permanent Manurial Experiment on Rice

The integrated application of green leaf manure (GLM) at 6.25 t ha^{-1}combined with Soil Test Crop Response (STCR) based N, P_2O_5, and K_2O, resulted in the highest grain yield of rice (6275 kg ha^{-1}) with a benefit-cost ratio (BCR) of 1:1.45. An increase in soil organic carbon was observed in all manured plots except the unmanured control.

7. Long-term STCR-IPNS experiment on Rice-Rice sequence

A 26-year-oldlong-termexperimentonRice-Ricesequence with STCR-IPNS technology recorded a built up in soil organic carbon (4.6 to 8.3 gkg^{-1}) and available P (20.2 to 31 kg ha^{-1}), a marginal decline in available K (670 to 563 kg ha^{-1}) and an increase in yield by 26 (6.85 t ha^{-1}) and 24.5 % (6.05 t ha^{-1}) in *kharif* and *rabi* seasons, respectively over blanket recommendation.

In a soil nutrient status with 28and 438kg $ha^{-1}$of available P and K, respectively, a fertilizer saving of26 kg of P_20_5 ha^{-1} and 25 kg of K_2O $ha^{-1}$was achieved owing to adoption of STCR-IPNS.

8. Comparative Metabolic Response of Rice Varieties to Salinity and Sodicity Stress

Alcohols, Sugars and derivatives, Alkanes, Alkenes, Diterpene, Methyl esters, Carboxylic acids and Polyamines were identified as metabolites.TRY 3 and TRY 5 showed an increase in sugars and derivatives, carboxylic acids and methyl esters under saline/sodic condition. A decrease in alkane was observed when compared to normal soil.

9. Enhancing the Nutrient Use Efficiency of TNAU Micronutrient Mixture through Vermicompost and Bio-inoculants for improving the Productivity of Rice in Sodic Soils

In an experiment to enhance productivity of Rice in Sodic Soils with TNAU Micronutrient Mixture in combination with vermicompost and Bio-inoculants, microbial populations were higher with combination of TNAU micronutrient mixtures (TNAU-MN) and vermicompost compared to combination with farmyard manure (FYM). The reduction in the ratio of micronutrient mixture to vermicompost (1:10, 1:7.5, 1:5, and 1:2.5) did not reduced the microbial populations. When micronutrients were applied in chelated form with organics such as FYM or vermicompost, their availability increased from the initial period and was maintained for up to 45 days. The application of bio-inoculants (solubilizers) along with organics further increased the availability of micronutrients in the soil throughout the incubation period. Also, the physical condition of the mixtures *viz.,* caking or colour change was not observed during the incubation period.

10. Development of Critical Level based Nutrient Management for Rice in Tamiraparani Command Area

Soil application of100% STCR-IPNS combined with root dipping and foliar application of deficient nutrients *viz.,* Cu, Zn and Mn, stage-specific bio-inoculants, and Zinc solubilizer, significantly produced a higher grain yield. However, the soil application of Cu, Zn and Mn alone registered a higher Benefit-Cost Ratio (BCR).

11. Dynamics of Metabolites and Microbes under different Organic Amendments in Rice

The incorporation of Navathania seeds exhibited a high mineralization rate, reducing the carbon to nitrogen ratio from 40.83 to 24.00 over 45 days of incubation with various organic sources. The incorporation of Navathania plant biomass @ 5.0 t ha^{-1}resulted in a higher grain yield (4950 kg ha^{-1}) and straw yield (7200 kg ha^{-1}) of rice compared to other treatments. An increase in grain yield ranging from 10.6 to 24.3% was recorded with the application of different organic sources.

12. Impact of Neem Coated Urea and Nano Urea on Rice under Tamiraparani river basin

Basal application of 25 % of N as urea + foliar spray of 1 % neem coated urea registered higher grain yield in ASD 16 and ADT 47 varieties of Rice with more BC ratio.

New Action Plans proposed for 2024-2025

Action Plan 1:Impact of Foliar Application of TNAU – WSF on Rice Yield

Objective

- To assess the effect of foliar application of TNAU- WSF on rice yield

Treatment Details

T_1– Absolute control
T_2– 100 % RDF
T_3– 25 % of basal + 1 % spray at 20, 35 & 50 DAT
T_4 – 50 % of basal + 1 % spray at 30, 45 & 60 DAT
T_5 – 75 % of basal + 1 % spray at 40, 55 & 70 DAT
T_6 – 25 % of basal + 2 % spray at 20, 35 & 50 DAT
T_7 – 50 % of basal + 2 % spray at 30, 45 & 60 DAT
T_8 – 75 % of basal + 2 % spray at 40, 55 & 70 DAT

Duration: One Year, **Design**: RBD, **Replications**:3

Observations to be recorded

1. Biometric observations
2. Yield and yield attributes
3. Soil available nutrients and nutrient uptake
4. Benefit Cost Ratio

Lead centre and Scientists	AC&RI, Madurai	Dr. S. Sheeba, Professor (SS&AC) Dr. J. Prabhaharan, Assoc. Prof. (SS&AC)
Co-ordinating centre & Scientists In-charge:	VOC AC&RI, Kilikulam	Dr. K. Baskar, Professor (SS&AC)
	ARS, Paramakudi	Dr. K. Muthuramu, Assoc. Professor (PBG)
	ADAC&RI, Trichy	Dr. M. Vijayakumar, Asst. Prof. (SS&AC) Dr. S. Rathika, Associate Professor (AGR)

Action Plan 2: Effect of various levels of Zinc with Zinc Solubilizing Bacteria (ZSB) to enhance Zinc availability and yield of Paddy

Objective

- To assess the effect of various levels of Zinc with Zinc Solubilizing Bacteria (ZSB) on the total zinc solubilisation and yield of paddy

Treatment Details

T_1– Absolute control
T_2– 100 % RDF
T_3 –T_2 + ZSB @ 500 ml ha^{-1}
T_4 – T_2 + $ZnSO_4$ @ 6.25 kg ha^{-1}
T_5 – T_2 + $ZnSO_4$ @ 12.5 kg ha^{-1}
T_6 – T_2 + $ZnSO_4$ @ 18.75 kg ha^{-1}
T_7 – T_2 + $ZnSO_4$ @ 25.0 kg ha^{-1}
T_8 – T_4 + ZSB @ 500 ml ha^{-1}
T_9 – T_5 + ZSB @ 500 ml ha^{-1}
T_{10} – T_6 + ZSB @ 500 ml ha^{-1}
T_{11} – T_7 + ZSB @ 500 ml ha^{-1}

Duration: One Year, **Design**: RBD, **Replications**: 3

Observations to be recorded

1. Biometric observations
2. Yield and yield attributes
3. Soil available nutrients and nutrient uptake
4. Benefit Cost Ratio

Lead centre and Scientists	AC&RI, Madurai	Dr. J. Prabhaharan, Associate Professor (SS&AC) Dr. K. Kumutha, Professor (AGM)
Co-ordinating centre & Scientists In-charge:	AC&RI, Killikulam	Dr. K. Manikandan, AP (SS&AC) Dr. Jebarlin Prabina, Professor (AGM)
	TRRI, Aduthurai	Dr. R. Bhuvaneswari, AP (SS&AC) Dr. P. Sivasankaridevi, AP (AGM)
	Dr.MSS AC&RI, Eachangkottai	Dr. R. Uma Sankareswari, Associate Professor (AGM), Dr. K. Suhathiya, Asst. Professor (SS&AC)

Project wise remarks				
S. No.	**Title**	**Name and Designation of the Project Leaders**	**Period**	**Remarks**
Action Plans				
1.	Management of multi-nutrient disorder in Rice	Dr. S. Jothimani, Professor& Head (SS&AC), Dept of SS&AC, VOC AC&RI, Killikulam **Co-ordinating centre** Dr. K. Manikandan, Asst. Prof. (SS&AC), TRRI, Aduthurai Dr. D. Janaki, Assoc. Prof. (SS&AC), ADAC&RI, Trichy Dr. M. Paramasivan Asst. Prof. (SS&AC), ARS, Thirupathisaram, Dr. S. Suresh, Professor and Head, ARS, Thirupathisaram	2022-2024	• Pooled mean may be worked out • Results may be given for information • To be closed
2	Effect of Fly Ash on improving Soil Properties and Rice Productivity	Dr. S. Sheeba, Professor & Head Dept. of Soils and Environment, AC&RI, Madurai **Co-ordinating centre** Dr. D. Lenin Raja, Asst. Prof, AC&RI, Killikulam Dr. G. Porkodi, Asst. Prof, SRS, Cuddalore	2023 – 2024	• Findings may be given for information • Silica uptake in different plant parts of rice may be assessed • To be continued
3.	Evaluation of Skipping of Phosphorus on Rice productivity in high P status soil	Dr. S. Sheeba, Professor & Head Dept. of Soils and Environment, AC&RI, Madurai **Co-ordinating Centre** Dr. D. Lenin Raja, Asst. Professor (SS&AC), AC&RI, Killikulam Dr. K. Manikandan, Assistant Professor (SS&AC), TRRI, Aduthurai	2022-2024	• Results may be given for adoption • To be closed
4.	Assessment of Nutrient Use Efficiency of TNAU – WSF in direct sown rice under drip irrigation	Dr. S. Sheeba, Prof. & Head, Dept. of Soils and Environment, AC&RI, Madurai Dr. J. Prabhaharan, Asst. Professor (SS&AC), AC&RI, Madurai **Co-ordinating Centre** Dr. K. Manikandan, Asst. Prof. (SS&AC), TRRI, Aduthurai	2023 – 2024	• Findings may be given for information. • New Action plan may be proposed to study the impact of foliar application of TNAU – WSF
On Farm Trial				
1.	Zinc nutrition for direct sown Rice through Zinc lysinate	Dr. J. Prabhaharan, Asst. Professor (SS&AC) **Coordinating Centres** Dr. K. Senthil, Prof. (Agrl. Chemicals), ADAC&RI, Trichy Dr. T. Balaji, Asst. Prof.	2023 – 2024	• Findings may be given for adoption • New OFT may be proposed

		(SS&AC), KVK, Ramnad Dr. S. Muthuramu, Assoc. Prof. & Head, ARS, Paramakudi Dr. K. Manikandan, Asst. Prof. (SS&AC), TRRI, Aduthurai Dr. T. Sivasankari Devi, Asst. Prof. (AGM), TRRI, Aduthurai		
University Research Projects				
1.	Permanent Manurial experiment in Rice based cropping system	Dr. K. Manikandan, Assistant Professor (SS&AC), TRRI, Aduthurai Dr. T. Sivasankari Devi, Assistant Professor (AGM), TRRI, Aduthurai	2022 – 2027	• Results may be given for information • To be continued
2.	Permanent Manurial Experiment on Rice	Dr. P. Christy Nirmala Mary Professor (SS&AC), Dept. of Soils and Environment, AC&RI, Madurai	2020 - 2025	• Results may be given for information • Heavy metal content in urban compost may be assessed • To be continued
3.	Evaluating potential Rice varieties suitable for saline and sodic soil conditions	Dr. A. Anuratha, Assoc. Prof. AC&RI, Keezhvelur Dr. A. Balasubramanian, Asst. Prof. (AGR), AC&RI, Keezhvelur **Co-ordinating centre** Saline condition – Dr. S. Vallalkannan, PC, KVK, Ramnad Sodic conditions – Dr. M. Baskar, Prof. & Head (SS&AC), ADAC&RI, Trichy Dr. S. Rathika, Assoc. Prof. (AGR), ADAC&RI, Trichy	2023-2025	• Findings may be given for information • To be continued
4.	Development of critical level based nutrient management for Rice in Tamiraparani command area	Dr. S. Jothimani, Professor and Head, Dept. of SS&AC, VOC AC&RI, Killikulam	Nov. 2020 to Oct. 2023	• Findings may be given for information • To be closed
5.	Enhancing the nutrient use efficiency of TNAU micronutrient mixture through vermicompost and bio-inoculants for improving the productivity of Rice in sodic soils	Dr. M. Baskar, Prof. & Head Dept. of SS&AC, ADAC & RI, Trichy Dr. S. Meena, Professor (SS & AC), TNAU, Coimbatore Dr. T. Uma Maheswari, Asst. Prof. (AGM), ADAC&RI, Trichy	April, 2023 to March, 2025	• Zn solubilizing bacterial count may be assessed • Project may be continued as per the technical programme

6.	Comparative metabolic responses of Rice varieties to salinity and sodicity stress	Dr. S. Meena, Professor (SS&AC), ADAC&RI, Trichy	March, 2022 – Feb, 2025	• Findings may be given for information • Project may be continued • ProposaL may be submitted for change of Project Leader
7.	Dynamics of metabolites and microbes under different organic amendments in Rice	Dr. D. Janaki, Associate Professor (SS&AC), KVK, Sirugamani Dr. K.G. Anitha, Professor (Ag. Micro.), ADAC&RI, Trichy	May,2022 to April,2024	• Findings may be given for information • Extension proposal may be submitted
AICRP Project				
1.	Long term STCR-IPNS Experiment on Rice-Rice sequence	Dr. S. Maragatham, Professor (SS&AC), Dept of SS&AC, TNAU, Coimbatore	Continuous experiment	• Results may be given for information and to be continued
External Funded Project				
1.	Impact of Neem coated urea and Nano urea on rice under Tamiraparani river basin	Dr. S. Jothimani Professor &Head (SS & AC), Dept. of SS&AC, VOC AC&RI, Killikulam	Nov. 2020 to Oct. 2023	• Results may be given for information • Extension proposal may be submitted

AGRICULTURAL MICROBIOLOGY

A1. Technology for Adoption : Nil

A2. Technology for OFT : Nil

A3. Technologies for Information:

1. Evaluation of the impact of novel synthetic microbial consortia on the growth and moisture stress alleviating potential in rice

Synthetic microbial consortia development for alleviating stress in rice resulted in standardizing optimum dose of different components of synthetic microbial consortia (*Sphingobium yanoiquiyae, Alkali halobacillus clausii* and *Azospirillum brasilense*) for seed treatment as $10^{3/}$ ml and AMF as 10 spores/seed through *in vitro* germination studies

2. **Harpin, a protein based smart formulations for defense priming and health in rice**

HpaG, a harpin is a multifunctional heat-stable protein secreted by the Gram-negative pathogen *Xanthomonas oryzae* pv. *oryzae* during pathogenesis. Primimg of rice seeds (ADT43) with this bacterial effector protein, HpaG improved the seed vigour, root length and bacterial blight tolerance. The metabolite analysis of root and leaf of the primed seedlings showed that there is 2.8-fold increase in IAA concentration. Further greater concentration of salicylic acid (10.75 ppm) over control seedlings (8.21 pppm) was evidence in primed seedlings indicating elicitation of SAR mediated defense.

3. **Development of AMF -based microbial inoculant package for aerobic/semi dry rice: Understanding the mechanism of action by metabolomic, proteomic and transcriptomic analyses**

Study on evaluation of abundance of AM fungi in rice ecosystem showed more abundance in rice- pulse/sugarcane/millet crop rotation systems compared to rice monoculture. Inorganically fertilized soil showed higher AM spore load and root colonization. Though the natural occurrence of *Rhizophagus intraradices* was found lesser in the rhizosphere of rice, the infectivity potential was found higher and recorded 60% infection in test plants (Maize).

4. Exploring and characterizing native probiotic strains in naturally fermented sour rice by MALDI-TOF for commercialization

Among the selected 4 varieties for fermented sour rice (Pazhayasadham) Mapillai samba is found to be best followed by White ponni, ADT 36 and CO 43 based on Sensory parameters, Physicochemical (pH, TA, colour value), Nutritional (CHO, protein, fat, fibre, minerals) and Functional parameters (antioxidant activity). Among the 12 LAB strains screened, 2 lactic isolates were known to possess best probiotic attributes (Acid tolerance – pH 2.0, Bile tolerance-0.5%, Cell surface hydrophobicity - xylene, exhibited no antibiotic tolerance).

5. **Development of Microbial consortium for drought mitigation of rice**

PPFM – *Methylobacterium aminovorans* Tm 13 and *Bacillus altitudinis* FD 48 are the standard PGPR bio-inoculants used for drought mitigation in rice. Hence both the efficient cultures are taken up for developing microbial consortium for drought mitigation in rice. Shelf-life study in liquid microbial consortium will be done with Individual cell concentration of 10^8 cells per ml. Shelf life of the two liquid biofertilizer consortia (PPFM and *B. altitudinis* FD 48) was prepared and tested for their individual population load at periodical intervals by serial dilution and plating method at periodic intervals up to 15 months. At present the consortia is 6 months old. The population was studied in monthly interval.

6. **Microbial synthesis of Zinc nanospheres to enhance Zinc use efficiency in rice**

Application of RDF + $ZnSO_4$ + zinc solubilizing bacteria performed on par with RDF + foliar spray of zinc nanoparticles with respect to yield. Application of zinc nanoparticles produced more antioxidant enzymes viz., peroxidase, poly phenol oxidase compared to the control plants

Remarks on the ongoing Action Plans /URPs/ Core/ AICRPs/ Externally Funded Projects

List of Projects reviewed: 6

Project details	Name and Designation of the Project leader	Duration	Project wise remarks
Action Plan			
Action plan no. Harpin, a protein based smart formulations for defense priming and health in rice.	Dr. U. Sivakumar, Prof. (Agrl. Micro.) Dr. R. Arul, Prof. (Biotechnology) Dr. C. Gopalakrishnan, Prof. (Plant Pathology),	2021-2023	Project may be continued as per approved technical program and results may be given for information
DST No: SPG/2022/000112 DST-SERB/ NRM/ AGM/ CBE/2022/R001Evaluation of the impact of novel synthetic microbial consortia on the growth and moisture stress alleviating potential in rice	**PI**: Dr T. Kalaiselvi, Professor (Microbiology) **Co-PIs**: Dr. E. Kokiladevi, Prof. & Head (Plant Biotech.) Dr. S.K. Natarajan, Assoc. Professor (Agronomy) Dr. M.R. Latha, Professor (SS&AC)	June 2022 to May 2025	The project may be Continued
University Research project-1			
NRM/CBE/AGM/RIC/2020/001. Exploring bioinoculants for developing coated seeds for enhanced nutrient uptake in Rice	**PI**: Dr. M. Gnanachitra Professor (Micro.) **Co-PI** (s): Dr. D. Balachandar Professor (Micro.) Dr. Jerlin, Prof. (SST)	Nov. 2019 to Oct. 2024	Due to crop failure, extension proposal may be submitted for one year.

NRM/ADT/AGM/RICE/ 2023/191 Development of Microbial consortium for drought mitigation of rice	**PI**: Dr. T. Sivasankari Devi, Asst. Prof. (AGM) TRRI, Aduthurai	Julv 2023 to June 2026	The project may be Continued. A common media may be developed for coculturing both the organism viz., PPFM – *Methylobacterium aminovorans* Tm 13 and *Bacillus altitudinis* FD 48.
NRM/TRY/AGM/RICE/ 2023/032 Exploring and characterizing native probiotic strains in naturally fermented sour rice by MALDI-TOF for commercialization	**PI:** Dr. T. Uma Maheswari, Asst. Prof. (Ag. Micro), Dr MSS AC&RI, Thanjavur **Co-PI:** Dr. K. Geetha, Professor (FSN), ADAC&RI, Trichy Dr. S. Geethanjali, Asst. Prof. (Biochem.), ADAC&RI, Trichy	Jan 2023 to Dec 2024	Project may be continued as per approved technical program and results may be given for information
Student Work			
Development of AMF - based microbial inoculant package for aerobic/semi dry rice: Understanding the mechanism of action by metabalomic, proteomic and transcriptomic analyses	Dr. U. Sivakumar Professor (Agrl. Microbiology) TNAU, CBE Dr. Akihiko Kamoshita Associate Professor Asian Research Centre for Bioresources and Environmental Sciences The Graduate School of Agricultural and Life Sciences, The University of Tokyo Student: Geethanjali. M (2020611001), Ph. D. in Agricultural Microbiology TNAU, Coimbatore – 03	April 2022 to March 2024	Findings may be given for Information

IV. CROP PROTECTION

AGRICULTURAL ENTOMOLOGY

A. Adoption/OFT/Information

I. Technology for Adoption

Adoption 1. Newer insecticide molecules on emerging pests in Rice

- Two sprays of chlorantraniliprole 18.5 % SC @ 150 ml/ha or diafenthiuron 50 WP @ 600 g/ha at 25 & 40 DAT reduced Whorl maggot, Hispa beetle & Black bug damage.
- Treated plots recorded highest yield (5513 kg/ha), CB ratio (2.24) & predators (1.41 nos./ plant).

Adoption 2. Efficacy of bio-products for the management of Earhead bug and Black bug of rice

1. *Beauveria bassiana* (1 x 10^8 cfu/g) @ 2.5 kg/ha - first spray during flowering & second a week later - reduced earhead bug & black bug damage.
   - Treated plots recorded highest yield (5046 kg/ha) & CB ratio (2.14)
   - Recommended as a better alternative to chemical application in organic rice cultivation.

II. Technology for OFT

OFT 1: Management of Earhead Bug in Rice

Treatments:

T1- Thiamethoxam 25 WG @ 100 g/ ha
T2- Azadirachtin 1% @ 1000 ml/ha
T3- Control (Untreated check)

Spray: Twice at Flowering & Grain filling stage
Layout: Randomized Block Design
Treatments: Three
Replications: Ten (Divide each plot (500 sq. m area) into 10 sub-plots of 50 square meter area).

Monitoring Scientist: Dr. V.A Vijayashanthi, Asst. Professor (Entomology), KVK, Tirur	
Centers to be involved	
KVK, TIRUR	Dr. V.A. Vijayashanthi, Asst. Professor (Entomology)
HC&RI (W), Trichy	Dr. S. Sheeba Joyce Roseleen, Asst. Professor (Entomology)
RRS, Paiyur	Dr. P. Govindan, Asst. Professor (Entomology)
AC &RI, Killikulam	Dr. L. Allwin, Assoc. Professor (Entomology)
AC &RI, Madurai	Dr. B. Usharani, Assoc. Professor (Entomology)
ARS, Bhavanisagar	Dr. V. Sathyaseelan, Asst. Professor (Entomology)

Observations to be recorded

1. Population of EHB at fortnight intervals/10 random hills/replication from flowering stage
2. Sweep net collection at weekly intervals (5 sweeps/ replication)

(1 sweep is a semicircular sweep for 180° to and fro)
3. Predators and parasitoids population / ten plants at random in each plot.
4. Yield (kg/plot)

OFT 2. Large scale demonstration of Residue free Rice Pest Management (Farmers field)

Treatments:
T1- Cowpea as border crop; light trap @ 1/ z (5%) followed by Azadirachtin 1% @ 1000ml/ha.

T2- Control (Untreated check)
Plot size -50 cents each
Treatments: Two

Replications: Ten (Divide each plot (2000 sq. m area) into 10 sub-plots of 200 sq.m area).

Monitoring Scientist: Dr. P. Anandhi, Assoc. Professor (Entomology), TRRI, ADT

Centers to be involved

	No. of trials*	
TRRI, ADT	4	Dr. P. Anandhi, Assoc. Professor (Entomology)
TNAU, CBE	4	Dr. SheelaVenugopal, Asst. Professor (Entomology)
MSSAC&RI, ECK	1	Dr. V. Mathirajan, Professor (Entomology)
AC&RI,Madurai	1	Dr. G. Srinivasan, Professor (Entomology)
HC&RI (W), Trichy	1	Dr. P. Yasodha, Assoc. Professor (Entomology
RRS, Paiyur	2	Dr. P. Govindan, Asst. Professor (Entomology)
VOC AC &RI, KKM	1	Dr. L. Allwin, Assoc. Professor (Entomology)
KVK, Sirugamani	1	Dr. SheebaJasmine, Asst. Professor (Entomology)
KVK, Tirur	4	Dr. V.A. Vijayashanthi, Asst. Professor (Entomology)

***Farmers field**

Observations: To be taken from 15 days after transplantation till pre-harvest stage at weekly interval on the existing pests and natural enemies as indicated below

1. Thrips damage % (leaf damage)
2. Stem borer damage % (DH/WE)
3. Leaffolder damage % (leaf damage)
4. Gall Midge damage % (silver shoot)
5. No of leaf/planthoppers/earhead bugs/10 hills
6.Any other pests (if noticed)
7. Natural enemies such as coccinellids, rove beetles and spiders/ 10 hills
8. Grain yield/plot
9. CB ratio

III. For Information

I. Monitoring of changing insect pest scenario in rice ecosystems

Assessment of pest and predator population through fixed plot, light trap, pheromone trap and roving survey were done at all the centres and the major pests and their natural enemies were correlated with the weather parameters.

a. Fixed plot survey

- The major pests noticed during samba season was stem borer (48th - 52nd standard week), leaffolder (47th -5th SW), earhead bug (50th SW) and whorl maggot (47th SW).
- Maximum temperature had significant negative correlation with earhead bug population (-0.5329**) and stem borer damage (-0.4576**) while hispa damage was significantly positively correlated (0.3906**).
- Minimum temperature had significant negative correlation with leaffolder damage (-0.6357**). Relative humidity was significantly negatively correlated with thrips (-0.4154**) damage. Sunshine was significantly positively correlated with hispa damage (0.5598**).
- Rice mite population was positively correlated with maximum and minimum temperatures and negatively correlated with the relative humidity (morning and evening) and rainfall. Regression equation was Y= -84.93+1.45 Max. T+0.88 Min.T-0.16 RH M-0.17 RH E-0.32RF; R^{2} - 0.381272

b. Light Trap/ Pheromone monitoring

- The adult yellow stem borer population was higher in March, gradually reduced in April and May, and again increased in October. Maximum BPH and GLH were caught in the second week of January and September, respectively.
- Rove beetle and coccinellid populations were high in March & August

c. Roving survey

- Maximum leaffolder damage was (15-17%) observed in Trichy district, during 52 - 62 DAT (^{4th} and ^{5th} week of December) on TPS 5 variety. Stem borer damage was maximum (10-11.5% dead heart) on 47 DAT (3rd – 4th week of December) and 88 DAT (10.5 % white ear) during *Thaladi* season.
- In Tiruvallur district, maximum dead heart incidence was observed during second week of July 2022 (12.50 %) and maximum leaf folder incidence was observed as 14.5% leaf damage during 3rd week of Sep 2023 and 11.5 % during 3rd week of Jan 2024.
- The highest GLH population was observed at Krishnagiri in November 2023 (5.68 nos/hill) and March 2024 (7.55 nos/hill). The BPH population was high in December 2023 (8.10 nos/hill) and March 2024 (8.25 nos/hill), and thrips occurred in October 2023 (15.50 nos/hill) and February 2024 (14.25 nos/hill).
- Leaf damage by leaf folder to an extent of 45 – 50 per cent was recorded in rice crop at flag leaf stage in BPT 5204 on 22.12.2023 at Karukkakottai village of Orathanadu block of Thanjavur district. BPH outbreak (>40%) was noticed at Mayiladuthurai, Sirkazhi and Thiruvidaimaruthur block during 1st week of Jan, 2024.

- Leaf folder damage was maximum during October, December and February (10-12%) at Madurai. The gall midge damage was maximum during September to December (10-12 %). During 2023-24, the damage by stem borer exceeded ETL in terms of white ear during January and February (3-4%). Thrips population was maximum (14 nos./sweep) during December.
- Stem borer species complex determination study conducted at Aduthurai, Coimbatore, Trichy and Ambasamudram in rice through light trap, pheromone trap and destructive sampling revealed that Yellow Stem borer (YSB), *Scirpophaga incertulas* as the predominant species and had a proportion of infestation of 80 per cent to 100 per cent in all the centers.
- YSB (< 5.0- 6.7 per cent incidence) and Dark Headed Borer (DHB), *Chilo polychrysus* (< 4.0- 18.2 per cent) was found in all the centres at vegetative stage. Until 45 days after planting, the crop was infested only by YSB and DHB. Pink Stem Borer (*Sesamia inferens*) infestation was found at the late tillering and panicle initiation until harvest.

II. Exploring insect resistance mechanisms

- Seven MLT cultures *viz*., AS 20026, ACM 15016, AS 19103, TNTRH 127, TKM 13, ACM 20044 and ADMS 20032-2 and one ART culture *viz*., TRY 5 were found to be moderately resistant to BPH with damage score <=5 at TNAU, Coimbatore while only one culture MLT 317 (CB 20164) was found to be moderately resistant at TRRI, Aduthurai under artificial screening.
- Seven MLT cultures were found to be moderately resistant to WBPH *viz*., MLT 105 (ACM 15016), MLT 203 (AS 20020), 204 (TNTRH 127), 214 (CB 18585), 217 (ACK 14090), 303 (TM 15127) and 410 (ADMS 20032-2) at TNAU, Coimbatore.
- Three cultures, MLT 105 (ACM 15016), MLT 204 (TNTRH 127) and MLT 410 (ADMS 20032-2) were found to be moderately resistant to Bph and WBph at TNAU, Coimbatore.
- Eighteen cultures *viz*., MLT 107 (TM 13377), 108 (AD 20020), 113 (AD 20020), 201 (CB 17537), 211 (TKM 13), 304 (AD19509), 403 (AD 20420), 404 (ADT 51), 405 (AD 20444), 406 (AD 20389), 501 (PM 19012), 502 (TKM 15), 508 (TNAU Pyr 01), 602 (CO54), 603 (TNAU Pyr 02#), 604 (CO 43 sub 1), 704 (AD 21205) and 801 (CB HTR 22001) showed less than 10% dead hearts and 2% white ears while the maximum infestation was 25 % dead hearts and 26.72 white ears in other cultures under field screening at four centres *viz*., Aduthurai, Ambasamudram, Coimbatore and Madurai.
- Two cultures *viz*., MLT 109 (ACK 13005) and 303 (TM 15127) showed < 2 % white ears but > 10% dead hearts.
- Nineteen cultures *viz*., MLT 110 (ADT 53), 111 (AS 19103), 112 (CB 18577), 121 (ADT 57), 208 (TM 15703), 213 (AD 20269), 305 (TPS 3), 311 (AD 18545), 312 (CB 19136), 319 (ADKKV 20002), 504 (CB HTR 22011), 506 (Co 53), 507 (CB 17561), 510 (AD 21096), 511 (TM 14022), 601(TNAU PYR 03*), 605 (TRY 5), 702 (CB 21102) and ART 12-1 (TRY5) showed < 10% dead heart and < 5% white ear.
- A new species of mite was observed during Sep-Oct 2023 which resembled *Schizotetranychus* sp. (web mite) with characteristic white rectangular webs on leaf lamina.

- Among the 30 entries screened, three entries *viz*., *Norungan, Kaligar* and *Kallurundaikar* showed minimum of <1% yellow stem borer incidence followed by 1-2% incidence in *Rathasali, Mattaikar, Neelan samba, Annam Azhagi, kuruvaikalangium, Thooyamalli, Nootripathu, Kadaikazhuthan, Kattuyanam,* and *Mullankaima.*
- Minimum incidence of leaf folder incidence (<1%) was recorded in six entries viz., *Neelan samba, Kallurundaikar, Nootripathi, Karrupukavuni, Sivappukavuni* and *Valan Samba* followed by 1-2% damage in *Rathasali, poongar, Norungan, Annam azhagi, Kaligar, Ottukitchili, Tooyamalli* and *Kattuyanam*.
- Out of 170 landraces field screened for stem borer resistance at Coimbatore, 15 showed zero dead hearts and zero white ears *viz*., *Ghandasala, Malayalathan Samba, Chithiraikar, Kullakar, Ghangai, Karuthakar, Karuppukavuni, Kattanam, Sengar, Salem-3, Sadai Samba, Rajamudi, Varigarudan Samba, Muthuvellai*, ADT-LR-75.
- Twelve landraces showed <10 % dead hearts and zero white ears *viz*., *Milagu Samba, Ottadaiyan, Kerala Pokkali, Chandikar, Chinnapunchai, Karimbalan, Kottanel, Onamuttan, Sivappu Chithiraikar, Mysore Malli, Karuppunel, Varisuriyan*.
- *Thengaipoo Samba* showed zero dead heart but 3.13 % white ear.
- Seventeen landraces *viz*., *Norungan, SivappuKavuni, Mattakuruvai, Katta Samba, Salem Channa, Red Sirumani, Poongar*, ADT LR-70*, Nootripathu, Haladichudi, Jhondi, Jothi, VasanaiJeeraga Samba, Arupatham Samba, Karnel, Manvilayan and VarappuKudaichan* showed <10 % deadheart and <5% white ear.
- *Shenmolagi, ThulasiVasanai Samba, Ponni, Kallimadaiyan, Anaikomban, Ara Samba, Thogai Samba and KattuPonni* showed > 25 % white ear and found to be susceptible under field screening.
- One landrace, *Mysore Malli* was found to be resistant and four land races *viz*. *Chetty Samba, Vellai Kombi Samba, Panamara Samba and VellaiGundu Samba* were found as moderately resistant to BPH through student thesis research work.
- *Mysore Malli* and *Vellai Kombi Samba* were least preferred by BPH nymphs showing higher antixenosis or non preference action against the pest.
- Feeding rate of BPH was significantly lower in PTB 33 (30.50 mm²) which was 13.39 times lower than the susceptible check TN 1 (408.50 mm2), followed by *Mysore Malli* (35.25 mm²), *Panamara Samba* (41.25 mm²), *Vellai Kombi Samba* (48.50 mm²), *Chetty Samba* (54.25 mm²) and *Vellai Gundu Samba* (68.75 mm²).
- PTB 33 recorded significantly lower amount of total sugars (3.94 mg/g) followed by *Mysore Malli* (4.94 mg/g) while TN 1 recorded 8.17 mg/g before infestation. The higher per cent reduction in protein content after BPH infestation was observed on TN1 (44.26 per cent) followed by *Vellai Kombi Samba* (32.99 %) and *Vellai Gundu Samba* (29.51 %). PTB 33 recorded the highest phenol content before (5.25 mg/g) and after (7.73 mg/g) infestation and TN 1 showed the least (2.48 and 3.25 mg/g, resp.). PTB 33 recorded the highest per cent increase in phenol content (47.44 %) followed by *Vellai Kombi Samba* (46.08 %) after BPH infestation.

- Out of the total 67 secondary metabolites identified, 21 metabolites were found to be upregulated, 12 metabolites got downregulated and 16 metabolites got differentially expressed 24 HAI. Most of the metabolites which got upregulated were fatty acids, organic acids, phenols and some sugars.
- Out of the 26 upregulated metabolites identified from samples taken 96 HAI, 23 metabolites got expressed only after infestation, of which around 11 were exclusively expressed in PTB 33. Twenty-one metabolites were downregulated and 10 were differentially expressed in 96 HAI samples.

III Pest Management

- Minimum mean mite population of 1.58 nos. / 10 cm leaf length was recorded in the plot which received the application of Spiromesifen 240 SC @ 0.8 ml/lit twice at 15 days interval followed by the plot treated with propargite 57 EC @ 2ml/lit twice at 15 days interval which recorded the mite population of 2.17 nos./ 10 cm leaf length.
- Thiamethoxam 25 WDG @ 100 g/ha was recorded to be superior in the management of earhead bug with the lowest population of 0.35 bugs per 10 hills after two sprays. Other chemical treatments *viz*., acephate 75 SP @ 1 kg/ha and chlorantraniliprole 18.5 SC @ 150 ml/ha were found to be next in the order of efficacy.
- After the first and second application of chlorantraniliprole18.5 % SC @ 150ml/ha, whorl maggot, hispa and black bug damage got significantly reduced and showed more number of spiders and coccinellids followed by diafenthiuron 50 WP @ 600g/ha.
- Among the two bio-agents, *Beauveria bassiana* was found to be superior to *Verticillium lecanii* in bringing down (70.07 %) the earhead bug population and both were significantly superior to the untreated check. *Beauveria bassiana* ($1x10^8$ CFU) @ 2.5 kg/ha, shall be recommended as a better alternative to chemical application in organic rice cultivation.
- Soil application of silicon granules @ 25 kg/ac at 15, 30 and 45 DAT along with foliar application of silicon powder 1% at 15, 30 and 45 DAT reduced the damage caused by yellow stem borer, attracted the parasitoids (*Telenomus* sp. and *Tetrastichus* sp.) and predators (dragonfly, damselfly and spider) and increased the growth and yield of rice.
- Silicon deposition in cell wall resulted in mandible damage and induced biochemical defences by triggering the defensive chemicals such as total phenols, and enzymes *viz*., PAL, PO and PPO.
- Fifteen indigenous *Bacillus thuringiensis* isolates, along with the reference strains (HD1 and 4AT1) were characterized and assessed for their toxicity against the rice leaffolder.
- The *Bacillus thuringiensis* isolates were PCR screened for lepidopteran toxic genes *viz*., cry1, cry2, cry9 and vip3A using gene specific primers. Three isolates *viz*., T29, T355 and T527 were positive for cry1Aa, cry1Ab, cry1Ac, cry2Aa, cry2Ab and vip3A genes. The isolate 543 had cry1Aa, cry1Ab, cry1Ac, cry2Aa and cry2Ab genes. The isolate T152 had only one gene, cry1Ac and T15 isolate had two genes namely, cry1Ac and cry2Aa. None of the isolates had cry9 gene.

- Leaf surface coated bioassay against the neonate larvae of rice leaffolder revealed that six isolates showed 100 per cent larval mortality which was on par with the reference strain HD1. The other isolates exhibited mortality ranged from 16.67 – 26.67 per cent.
- *Beauveria bassiana* (Bb 3) isolates were collected from Needamangalam, Thiruvarur District recorded 80 per cent mortality under laboratory condition against rice leaffolder. Bb3 talc-based isolates showed 44.88 per cent reduction over control against rice leaffolder under green house condition. The talc-based formulation (Bb3 isolates) @ 3 kg/ha recorded lowest mean damage of leaffolder (6.25) with a per cent reduction of 47.18 over control under field conditions.
- Clove buds extract of 0.2% was found to be effective against the development of lesser grain borer, *Rhyzopertha dominica* in stored paddy by reducing 95.2% infestation.
- Clove oil-based treatment of 5% dust formulation @ 4g/100g to stored unhusked paddy grains was effective in reducing the infestation up to 92.46 % of the lesser grain borer, *Rhyzopertha dominica* effecting a weight loss benefit ratio of 2.57.
- Based on acute toxicity studies, flupyradifurone was found to be toxic to Angoumois grain moth, *Sitotroga cerealella* followed by spinetoram.
- Spinetoram was found to be toxic followed by emamectin benzoate against lesser grain borer, *Rhizopertha dominica*.
- In the studies on efficacy of ultrasonic sound on repellence of Rice Weevil (*Sitophilus oryzae*) in stored grains, Ultrasonic Transmitter circuit has been designed to deliver optimum resonance frequency of 115 – 172 KHz which is expected to have maximum repellence of Rice Weevil.

B. Action plan (2024-2025)

1. Agricultural Entomology

Theme Areas:

S. No.	Agricultural Entomology
1.	Monitoring of changing insect pest scenario in rice ecosystems
2.	Exploring insect resistance mechanisms
3.	Emerging Pest Management
4.	Residue free Pest Management

Action Plan 1: Monitoring of changing insect pest scenario in rice ecosystems

<table>
<tr><td>Theme leader</td><td colspan="3">Dr. R. Nalini, Professor (Entomology) Coordination at State level-AC & RI, MDU*</td></tr>
<tr><td>Activity</td><td>Name of the Scientist and Centre</td><td>Observations to be recorded</td><td>Deliverables/ expected out come</td></tr>
<tr><td>• Monitoring pest status and keeping vigilance on emerging pests
• Assessment of insect pest, mites and natural enemies population in situ.
• Impact of light trap on non-target arthropods wherever available
• Fixed plot survey at weekly interval in all the identified centres
• Roving survey at fortnightly interval in all the identified districts during the district specific</td><td>
<table>
<tr><th>Zone</th><th>Roving survey</th><th>Fixed plot survey</th></tr>
<tr><td colspan="3">North</td></tr>
<tr><td>KVK, Tirur</td><td>√</td><td>√</td></tr>
<tr><td colspan="3">Dr. R.A. Vijayashanthi, Asst. Professor (Ento.)</td></tr>
<tr><td>RRS, Paiyur</td><td></td><td>√</td></tr>
<tr><td colspan="3">Dr. K. Govindan, Asst. Professor (Ento.)</td></tr>
<tr><td colspan="3">West</td></tr>
<tr><td>TNAU, CBE</td><td></td><td>√</td></tr>
<tr><td colspan="3">Dr. Sheela Venugopal, Asst. Professor (Ento.)</td></tr>
<tr><td>ARS, BSR</td><td>√</td><td></td></tr>
<tr><td colspan="3">Dr. V. Sathyaseelan Asst. Professor (Ento.)</td></tr>
</table>
</td><td>• Fixed and roving survey on pest incidence.
• Recording of weather parameters.
• Multiple correlation and regression analysis with previous years weather data and pests damage and population data

* will consolidate the monthly data, make assessment of</td><td>Forewarning on emerging pests and pest management decisions
Intervention with suitable IPM package
Monthly documentation of pest status in the major tracts</td></tr>
</table>

crop seasons • Uploading of data in Google forms	**Cauvery Delta/East**			the pest scenario and submit state report o/b 25th of the month to the Department.	
	TRRI, ADT		√		
	Dr. P. Anandhi, Assoc. Professor (Ento.)				
	KVK, Sirugamani	√			
	Dr. Sheeba Jasmine, Asst. Professor (Ento.)				
	AC&RI, ECK	√			
	Dr. V.G. Mathirajan, Professor (Ento.)				
	South				
	AC&RI, MDU	√			
	Dr. R. Nalini, Professor (Ento.)				
	VOC AC&RI, KKM		√		
	Dr. L. Allwin, Assoc. Professor (Ento.)				
	KVK, Ramnad	√			
	Dr. J. Ramkumar, Assoc. Professor (Ento.)				

Action Plan 2. Identification of resistant sources and exploring insect resistance mechanisms

Theme Leader	Dr. Sheela Venugopal, Asst. Professor (Entomology), TNAU, Coimbatore		
Activity	**Name of the Scientist(s) and Centre(s)**	**Observations to be recorded**	**Deliverables**
Identification of resistance sources to major insect pests. Screening of TNAU (MLT/ART) entries Screening of local germplasms • Artificial screening for sucking pests: Standard Seedling Box Method of screening to be followed (CBE & ADT)	**TNAU, CBE** Dr. Sheela Venugopal, Asst. Prof. (Ento.) SB, BPH, WBPH (Field & Artificial Screening & Mechanism of resistance) **TRRI, ADT**	Levels of resistance in field screening/artificial as per standard procedure. Biophysical and biochemical characters of best pipeline entries only.	Resistant sources for BPH/WBPH/GLH/ Stemborer/Leaf folder. Underlying mechanism

• Field screening for stem borer & leaffolder (to be planned coinciding with peak pest incidence period of the respective centre or to be inoculated with insects or egg masses in case of low incidence). • Investigation of resistance mechanisms- Attributes contributing to antixenosis & antibiosis have to be evaluated for the best entries selected for Stemborer, Leaf folder and Sucking pests (CBE & ADT). (Protocol will be sent by the Theme Leader to the participating Scientists)	Dr. P. Anandhi, Asst. Professor (Entomology), SB, LF, BPH, GM (Field & Artificial Screening) AC&RI, MDU Dr. B. Usharani, Assoc. Professor, (Entomology), SB, LF (Field Screening) **VOC AC & RI, KKM (Hot spot)** Dr. L. Allwin, Assoc Professor (Ento.) SB, LF (Field Screening)		

Action Plan 3. Monitoring and management of leaf mite (*Oligonychus* spp.) and web mite, *Schizotetranychus* sp. (New)

Theme Leader	**Dr. E. Sumathi, Prof. (Entomology), TNAU, Coimbatore**		
Activity	**Name of the Scientist(s) and Centre(s)**	**Observations to be recorded**	**Deliverables**
Activity 1: Monitoring of *Schizotetranychus* sp. Activity 2: Management of rice web mite Need based application of T1-Spiromesifen 22.9 SC @ 0.8 ml/l twice at 15 days interval (DI) T2-Spiromesifen 22.9 SC@1.2 ml/l twice at 15 DI T3-Fenazaquin 10 EC@2 ml/l twice at 15 DI T4-Azadirachtin 1500ppm @ 5 ml/lit. or 10000 ppm@3ml/l twice at 15 DI T5. Control	TNAU, CBE Dr. E. Sumathi, Prof. (Ento.) TRRI, ADT Dr. P. Anandhi, Assoc. Prof. (Ento.) VOC AC &RI, KKM Dr. K. Elanchezhyian, Assoc. Prof. (Ento.) RRS, Paiyur Dr. K. Govindan, Asst. Professor (Ento.)	Survey has to be made in different parts of the district, collect mite samples and send to Dept. of Agrl. Entomology, TNAU. Population of mite from top, middle and bottom leaves (1x10 cm leaf length) at weekly intervals from the date when mite incidence is noticed. Yield and BCR	Identification of mite species Effective molecule for rice web mite management will be identified.
Replications :4; Design: RBD			

Action Plan 4. Documentation of natural egg parasitoids of YSB (New)
Theme Leader: Dr. Sheela Venugopal, Asst Professor (Agrl. Entomology)

10 egg masses to be collected every fortnight during the cropping seasons and the egg masses to be kept in glass tubes for observation of emergence of parasitoids. Then the species and number of parasitoids emerging from each egg mass to be observed and documented as follows

Theme Leader	: Dr. Sheela Venugopal, AP (Entomology) TNAU, CBE		
Activity	**Name of the Scientist(s) and Centre(s)**	**Observations to be recorded**	**Deliverables**
Activity 1: Collection of YSB egg masses **10 egg masses to be collected every fortnight during the cropping seasons and the egg masses to be kept in glass tubes for observation of emergence of parasitoids** Activity 2: Documentation of parasitoids The species and number of parasitoids emerging from each egg mass to be observed and documented as follows	TNAU, CBE Dr. Sheela Venugopal, AP (Agrl. Entomology) TRRI, ADT Dr. P. Anandhi, Assoc. Prof. (Entomology) VOC AC &RI, KKM Dr. K. Elanchezhyian, Assoc. Prof. (Ento.) KVK, Tirur Dr. R.A. Vijayashanthi, Asst. Prof. (Ento.)	No. of larva emerging from each egg mass No. of parasitoids emerging from each egg mass Identification of the parasitoid species *Trichogramma/ Tetrastichus/ Telenomus* or others	Natural parasitism in rice ecosystem could be determined Scope of mass culturing the promising species as biocontrol agent could be explored

Date	Egg mass 1			Egg mass 2			…	Egg mass 10		
	No. of larva	No. of Parasitoids	Species	No. of larva	No. of Parasitoids	Species		No. of larva	No. of Parasitoids	Species
15DAT										
….										
End of season										

C. Remarks on the ongoing URP/AICRIP/EPF/UCF
Total Number of Projects in Crop Protection

List of URP/AICRP/EPF/UCF Discipline	URP	AICRIP	EFP	UCP	Total
Agricultural Entomology	8	2	5	-	15

D. Remarks on the ongoing URP/AICRIP

S. No.	Project No. and Title	Remarks
1.	**CPPS/MDU/AEN/RICE/2023/276** Influence of silica and potassium solubilizing bacteria in inducing resistance against insect pests of rice. Dr. R. Nalini, Professor and Head, Dept. of Agric. Entomology, AC&RI, Madurai (October, 2023 - October, 2025)	The project may be continued
2.	**CPPS/MDU/ENT/RIC/2021/001** Screening of certain botanicals for the management of lesser grain borer, *Rhyzoperthadominica* (Coleoptera: Bostrychidae) in stored paddy grains. Dr. J. Jayaraj, Professor (Agrl. Ento.), Dept. of Agricultural Ento., AC&RI, Madurai (March 2021 – March 2024)	The project may be closed
3.	**CPPS/TIR/AEN/RICE/2023/102** Evaluation Of Traditional Paddy Varieties Against Major Insect Pests PI: Dr. V.A. Vijayashanthi, Asst. Professor (Agrl. Entomology), KVK, Tirur CO-PI: Dr. P. Yogameenakshi, Associate Professor (PBG) RRS, Tirur)- 2023-2024	The project may be continued
4.	**CPPS/CBE/AEN/RICE/2023/222** Management of major insect pests in stored paddy Dr. G. Preetha, Associate Professor (Agrl. Entomology), Directorate of Seed Centre, TNAU, Coimbatore (October 2023 to September 2025)	The Project may be continued
5.	**CPPS/CBE/AEN/RICE/2023/220** Screening of rice landraces for stem borer resistance and deciphering the mechanism of resistance. 1. Dr. Sheela Venugopal, AP, (Agrl. Ento.), Dept. of Rice, TNAU, CBE 2. Dr. R. Suresh, Assoc. Professor (PBG), Dept. of Rice Sep 2023-Aug 2025	The Project may be continued

6.	**CPPS/ ADT/ AEN/ RICE/ 2023/ 196** Pest weather relationship and weather-based pest forewarning model for the stem borer, BPH, GLH, Black bug and beneficial insects of Rice PI: Dr. P. Anandhi, Associate Professor (Agrl. Entomology), TRRI, Aduthurai- 612 101. CO PI- Dr. Patil Santosh Ganapati, Asst. Professor (Agri. Statistics), Department of Physical sciences & Information Technology, AEC&RI, TNAU, Coimbatore – 641 003. July 2023-June 2025	The Project may be continued
7.	**CPPS/TRY/CPS/RIC/2021/001.** Bio- ecology and management of Pink Stem Borer, *Sesamiainferens* Walker (Lepidoptera: Noctuidae) in rice Dr. Sheeba Joyce Rosleen, AP (Ento.) August 2021- July 2024	The Project may be continued
8.	**AEC&RI/CBE/RIC/2023/001** Efficacy of ultrasonic sound on repellence of rice weevil (*Sitophilus oryzae*) in stored food grains (Rice and pulses) Dr. Devaraj Soundarajan, Asst. Professor (Physics), AEC&RI, TNAU, Coimbatore Dr. Sheela Venugopal, Asst. Professor (Ento.), Dept. of Agrl. Entomology, TNAU, CBE (Mar 2023-Feb 2025)	The Project may be continued
9.	**AICRP/PBG/CBE/RIC/003:** All India Co-ordinated Rice Improvement Project – Entomology Part, (Coimbatore) Dr. Sheela Venugopal, Assistant Professor (Agrl. Ento.)	Project may be continued
10.	**AICRP/PBG/ADT/RIC/002** All India Co-ordinated Rice Improvement Project – Entomology Part (Aduthurai) Dr. P. Anandhi, Asso. Professor (Ento.)	Project may be continued

PLANT PATHOLOGY

A. Technologies for Adoption/OFT/Information

I. Technology for Adoption: --

II. Technology for OFT

OFT 1: Management of rice diseases through new molecules

Treatments

T_1 - Foliar spraying of Azoxystrobin 18.2% W/W + Difenoconazole 11.4% W/W SC @ 0. % at the time of symptom appearance

T_2 - Foliar spraying of Zineb 68% +Hexaconazole 4% WP @ 0.25 % at the time c symptom appearance

T_3 - Untreated control

Design : RBD
Replications : 7
Season : *Rabi*

Centres to be involved:

Dr. M.S.S. AC&RI, Eachangkottai [MS]*	**:**	**Dr. A. Ramanathan, Professor and Head (Plant Pathology)**
RRS, Ambasamudram	:	Dr. N. Rajinimala, Asst. Prof. (Plant Pathology)
AD AC &RI, Trichy	:	Dr. S. Thiruvudai Nambi, Professor (Plant Pathology)
AC & RI, Keelvelur	:	Dr. P. Ahila Devi, Asst. Prof. (Plant Pathology)

*** MS-Monitoring Scientist**

Observations: Severity of rice diseases, yield and CB ratio

OFT 2: Forewarning of rice fungal diseases through spore trapping technology

Treatments

T_1 - Inoculum cum weather based application of Azoxystrobin + Tricyclazole 1 ml/l two sprays @ 15 days interval

T_2 - Farmer practice (Curative application of Azoxystrobin + Tricyclazole 1 ml/l) three sprays @ 15 days interval)

T_3 - Untreated control

Design : RBD
Replications : 7
Season : *Rabi*

Centres to be involved:

Dept. of Plant Pathology, TNAU, Coimbatore [MS]*	**:**	**Dr. A. Kamalakannan, Prof. (Plant Pathology)**
Dept. of Rice, TNAU, CBE	:	Dr. C. Gopalakrishnan, Prof. (Plant Pathology)
RRS, Ambasamudram	:	Dr. N. Rajinimala, Asst. Prof. (Plant Pathology)
TRRI, Aduthurai	:	Dr. K. Rajappan, Professor (Plant Pathology)

*** MS-Monitoring Scientist**

Treatment spraying: Two sprays @ 15 days interval

III. For Information

- During *rabi* season of 2023 blast, brown spot, sheath blight, bacterial leaf blight, false smut and grain discoloration diseases were noticed in different parts of Tamil Nadu.

- The MLT entries *viz*., MLT 403, 404, 410, 603, 707 and 902 were resistant to BLB, sheath blight, sheath rot and brown spot.
- Minimum temperature and relative humidity in the evening were positively correlated with the development of blast disease while maximum temperature, wind velocity and evaporation were negatively correlated.
- Foliar spraying of propiconazole @ 25% EC @ 500 ml/ha recorded the minimum sheath blight disease incidence of 8.46 PDI followed by seed treatment with *Bacillus subtilis* (Bbv57) @ 10g/kg and soil application @ 2.5 kg/ha.
- Foliar spray of copper hydroxide @ 500 gm/acre recorded the lowest BLB incidence (12.7 PDI) and followed by foliar spray of bactrinashak @ 500 ppm/acre. Bactrinashak can be used as an alternative to fungicide.
- Seed treatment (10 g/kg) and two foliar spray (1%) of *B. subtilis* (CSB23) (I^{st} at boot leaf & IInd at 50% flowering stage) was found effective in reducing the false smut incidence besides increasing grain yield.
- 25 biomolecules were identified from resistant donor RG 170. Molecular docking was performed at TNAU, Coimbatore for the identification of mode of action and fungicidal activity of biomolecules against false smut pathogen, *Ustilaginoidea virens*.
- The biomolecule, pyroglutamylvaline had the highest binding energy to virulent protein target site of *U. virens* namely, chitin synthase (- 6.6 kcal/mol.) and low affinity iron transport protein UVT 3277 (- 7.8 kcal/mol.).
- The binding energy was comparatively less in propiconazole at the above two sites (- 3.5 and - 3.2 kcal/mol. respectively) which was used as positive control for assessing the fungicidal activity.
- Rice plants treated with *B. amyloliquefaceins* (B44) (ON624362) altered the expression of transcription factor (WRKY45), wall associated kinases 85 (WAK 85), chitin elicitor receptor kinase 1 (CERK 1), lipoxygenases (LOX), jazmonate zim domain protein (JAZ), receptor like kinase for flagellin (FLS 2), Peroxidase (PO), polyphenol oxidase (PPO), phenylalanine ammonia lyase (PAL) and pathogenesis related protein 1 (PR 1) genes responsible for plant defense.
- Copper hydroxide, propiconazole, trifloxystrobin + tebuconazole and azoxystrobin + propiconazole @ 1000 ppm concentration were found to be very effective with 100% inhibition under *in vitro* conditions.
- The lowest false smut incidence of 19.5 PDI was recorded in the treatment applied with 100:100:125% N, P_2O_5, K_2O fertilizers and sprayed with trifloxystrobin + tebuconazole 0.05% under field conditions.

B. Action plan (2024-2025)

Theme Area

1. Disease monitoring, surveillance, epidemiological studies and artificial intelligence-based diagnosis
2. Optimization of artificial inoculation and management of false smut
3. Management of sheath blight with chemicals and bio-agents
4. Management of Bacterial Leaf Blight of rice

Action Plan 1: Disease monitoring, surveillance, epidemiological studies and artificial intelligence-based diagnosis (Cont.)

Theme Leader	Dr. K. Rajappan, TRRI, Aduthurai		
Activity	**Name of the Scientist and Centre**	**Observations to be recorded**	**Deliverables/ out come**
I. Monitoring of pest and diseases under irrigated and direct sown rice (blast, sheath blight, sheath rot, bacterial blight, brown spot, grain discoloration and false smut) diseases 1. Fixed plot survey 2. Roving survey 3. Correlation with weather factors. (All centers) II. Development of forecasting model for major diseases of rice using the available data in collaboration with ACRC. III. Artificial intelligence-based diagnosis	Dr. K.Rajappan, Professor (PP), TRRI, Aduthurai Dr. C. Gopalakrishnan, Professor (PP), Dept. of Rice, Coimbatore Dr. N. Revathy, Professor (PP), AC&RI, Madurai, Dr. J. Sheela, Professor (PP), AC&RI, Killikulam Dr. S. Thiruvudai Nambi, Professor (PP), ADAC&RI, Trichy Dr. N. Rajinimala, Associate Professor (PP), RRS, Ambasamudram Dr. V. Ramamoorthy, Professor (PP) Dr. M.S.S. AC &RI, Eachangkottai Dr. K. Chitra, Associate Professor (PP), KVK, Virunjipuram Dr. S. Sudhasha, Assistant Professor (PP), RRS, Tirur Dr. Patil Santosh Ganapati, Asst. Prof. (Agrl. Statistics), Dept. of Physical Sciences, TNAU, Coimbatore	Per cent disease incidence / PDI as per standard grades. Correlation and regression analysis of pest and disease progression during cropping periods in relation to weather parameters. Regular bulletins on pest and disease scenario in the particular zone should be given for the benefit of farmers through press and media marking a copy to Director (CPPS). Approximately 3000 or more images to be captured for each disease and centre Labeling and classification of symptoms	Timely monitoring of disease epidemics and fore-warning of farmers and line departments. Development of mobile app through AI

Action Plan 2: Optimization of artificial inoculation and management of false smut

Theme Leader:	Dr. C. Gopalakrishnan, Professor, Dept. of Rice, TNAU, Coimbatore		
Activity	**Name of the Scientist and Centre**	**Observations to be recorded**	**Deliverables**
Standardization of artificial inoculation methodologies Sequential inoculation of *U. virens* @ 10^8 spores / ml starting from initiation of flowering to completion of flowering. **Management of false smut** Foliar spray of Propiconazole @ 200 ml/ac at initiation of flowering and at 50% flowering Foliar spray of effective bacterial endophyte @ 1kg/ac at initiation of flowering and at 50% flowering Control Season: *Rabi* Replication: 7	Dr. C. Gopalakrishnan, Professor, Dept. of Rice, TNAU, Coimbatore Dr. K. Rajappan, Prof. (PP), TRRI, Aduthurai Dr. N. Revathy, Prof. (PP), AC&RI, Madurai Dr. N. Rajinimala, Associate Professor (PP), RRS, Ambasamudram Dr. S. Thiruvudai Nambi, Professor (PP), ADAC&RI, Trichy Dr. V. Ramamoorthy, Associate Prof. (PP), AC &RI, Eachangkottai	Standardization of inoculum Per cent disease incidence	• An artificial inoculation method will be available • Management technology

Action Plan 3: Management of sheath blight with chemicals and bio-agents

Theme Leader:	**Dr. P. Mareeswari, Professor (Pl. Path.) AC&RI, Madurai**		
Activity	**Name of the Scientist and Centre**	**Observations to be recorded**	**Deliverables**
Management of sheath blight T_1 - Foliar spraying of Bioagent @ 1 kg/ac T_2 - Foliar spraying of Propiconazole @ 200 ml/ac. T_3 - Control • First spray at the time of initial appearance of the disease and	Dr. P. Mareeswari, Professor (Pl. Path.), AC&RI, Madurai Dr. K. Rajappan, Professor (PP), TRRI, Aduthurai Dr. A. Ramanathan, Professor (PP), Dr.M.S.S. AC &RI, Eachangkottai Dr. M. Thamaraiselvi, Assistant	1. PDI at 15 and 30 Days after application 2. Yield (kg/ ha) 3. C:B ratio	To develop suitable method for rice sheath blight management

second spray at 15 days later Season: *Rabi* Replication: 7	Professor (PP), AEC &RI, Kumulur Dr. S. Sudhasha, RRS, Tirur		

Action Plan 4: Management of Bacterial Leaf Blight of rice

Theme Leader:	**Dr. N. Rajinimala, RRS, Ambasamudram**		
Activity	**Name of the Scientist and Centre**	**Observations to be recorded**	**Deliverables**
Management of BLB T_1 - Foliar spraying of bactrinashak @500 ppm T_2 - Foliar spraying of Copper hydroxide @ 5 ml/lit T_3- Foliar spraying of *Bacillus subtilis* (Bbv57) @ 1 kg/ac. T_4-Control • First spray at the time of initial appearance of the disease and second spray at 15 days later Season: *Rabi* Replication: 5	Dr. N. Rajinimala, Associate Professor (PP), RRS, Ambasamudram Dr. K. Rajappan,Professor (PP) Aduthurai Dr. C. Gopalakrishnan, Professor (PP), TNAU, Coimbatore Dr. P. Mareeswari, Professor (Pl. Path.), AC&RI, Madurai	Disease intensity and grain yield.	Identification of new chemical / biomolecule for BLB management.

C. Remarks on the ongoing URP/AICRIP/EFP/UCP *etc*.

Crop	Pl. Path. (Nos.)
University Research Projects	9
AICRP Projects	2
Externally Funded Projects	3
Total	**14**

2. PLANT PATHOLOGY

S. No.	Project No. and Title	Remarks
	University Research Sub Project	
1	**CPPS/TRY/PAT/RIC/2020/001:** Bio-intensive management of blast and bacterial leaf blight in paddy using *Streptomyces* spp. under salt affected soils (September 2020- August 2023) **Dr. V.K. Satya**	To be continued
2	**CPPS/KKM/PAT/RIC/2020/001:** Bio-control potential of spermosphere microbiomes on rice brown spot pathogen Bipolaris oryzae (Breda de Haan) Shoemaker (April, 2020 to March, 2023) **Dr. J. Sheela**	To be closed and a new project may be proposed in rice
3	**CPPS/KKM/PAT/RIC/2020/002:** Management of Bacterial Leaf Blight of Rice caused by *Xanthomonas oryzae* pv. *oryzae* using plant products and essential oils. (July 2020- June 2023) **Dr. N. Rajinimala**	To be closed and a new project may be proposed
4	**CPPS/TRY/PP/RIC/2002/002:** Efficacy of Panchakavya on the Management of Foliar diseases of paddy in saline soil eco system. (June 2022 - December 2025) **Dr. K. Sethuraman**	To be continued
5	**CPPS/TRY/PP/RIC/2022/001**: Studies on effect of weather parameters on the false smut (*Ustilaginoidea virens* (Cooke) Takahashi) disease incidence in rice (Oryza sativa L.) for developing forewarning model. (February 2022 - January 2025) **Dr. M. Rajesh**	To be continued
6	**CPPS/TRY/PATH/RIC/2023/001:** Management of rice sheath blight incited by *Rhizoctonia solani* khun using seaweeds and bio-agents. (March 2023 – February 2026) **Dr. A. Sangeetha**	To be continued
7.	**CPPS/ECK/PAT/RIC/2021/001:** Studies on epidemiology and management of false smut disease of rice caused by *Ustilaginoidea virens.* (August 2021- July 2024) **Dr. S. Mathiyazhagan**	To be continued
8.	**New:** Development of disease predictive model based on epidemiological factors for management of bacterial leaf blight of rice. (June 2022 – May 2025) **Dr. N. Revathy**	To be continued
9.	**CPPS/CBE/PAT/RIC/2019/001:** Management of aflatoxin contamination in rice through botanicals (July 2019 – June 2023) **Dr. C. Gopalakrishnan**	To be closed and a new project may be proposed in rice

	ICAR-AICRIP	
1.	**AICRP/PBG/CBE/RIC/003:** All India Co-ordinated Rice Improvement Project-Pathology part (Coimbatore) **Dr. C. Gopalakrishnan**	The research work may be continued as per the technical programme of AICRIP.
2.	**AICRP/PBG/ADT/RIC/002:** All India Co-ordinated Rice Improvement Project-Pathology part **Dr. K. Rajappan**	The research work may be continued as per the technical programme of AICRIP.

V. REMARKS

a. General remarks

- Theme based research action plans in rice may be formulated (**Action**: All Directors/Deans)
- Mechanization in rice cultivation may be popularized (**Action**: All Scientists/DEE).
- All the scientists may be encouraged to submit research proposals involving various disciplines for externally funding. (**Action**: All Scientists)
- All scientists may be encouraged to publish research articles in peer-reviewed journals with NAAS ratings more than 7.0 (**Action**: All Scientists)

b. Crop Improvement

- Development of herbicide tolerant rice varieties may be given priority (**Action**: DCPMB&B/DCPBG)
- Efforts may be taken to develop superfine rice varieties to replace BPT 5204 and RNR 15048 (**Action**: DCPBG)
- Suitable varieties may be identified for direct seeded condition (**Action**: DCPBG/DCM/DNRM).
- Efforts may be taken to develop improved landraces (**Action**: DCPBG).
- Seeds of all landraces available in all the research stations may be deposited in Ramiah Gene Bank (**Action**: DCPBG).
- Research on development of climate smart varieties may be intensified (**Action**: DCPBG/DCM).
- Efforts may be taken to develop therapeutic and nutritionally rich rice varieties (**Action**: DCPBG/DCPMB&B)
- Research on development of Low GI rice varieties may be strengthened (**Action**: DCPBG/Dean, CSC&RI, Madurai)
- Mass production and supply of quality seeds may be ensured (**Action**: Director, Seed Centre/DCPBG)

c. Crop Management

- Research on super nano urea may be initiated for improving crop productivity and nutrient use efficiency (**Action**: DNRM/DCM).
- Newer herbicide molecules may be tried to control emerging weeds in dry direct seeded rice system (**Action**: DNRM/DCM).
- Research on Direct Seeded Rice may be strengthened and suitable package of practices for DSR may be developed (**Action**: DCM/DNRM).
- Efforts may be taken to popularize rice crop boosters (**Action**: DCM/DNRM).

d. Crop Protection

- Biocontrol agent *Trichogramma* sp., botanical insecticides and insect monitoring tools *viz*., light trap, pheromone traps *etc*., may be effectively

utilized and large-scale demonstrations may be organized in farmers field (**Action**: DCPPS/DEE).

- Research on genome sequencing for pests and diseases resistance in land races may be taken up (**Action**: DCPMB&B/DCPPS).
- Systemic acaricides may be identified for the emerging mite species (**Action**: DCPPS).
- Seed treatment with clove oil formulations may be validated for storage pests (**Action**: DCPPS).
- Efforts may be taken to address the infestation of Nematode in direct seeded rice cultivation (**Action**: DCPPS).

V. List of participants

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6.	Dr. P. Balasubramaniam	Director, NRM, TNAU, Coimbatore
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